

μGlyph: a Microgesture Notation

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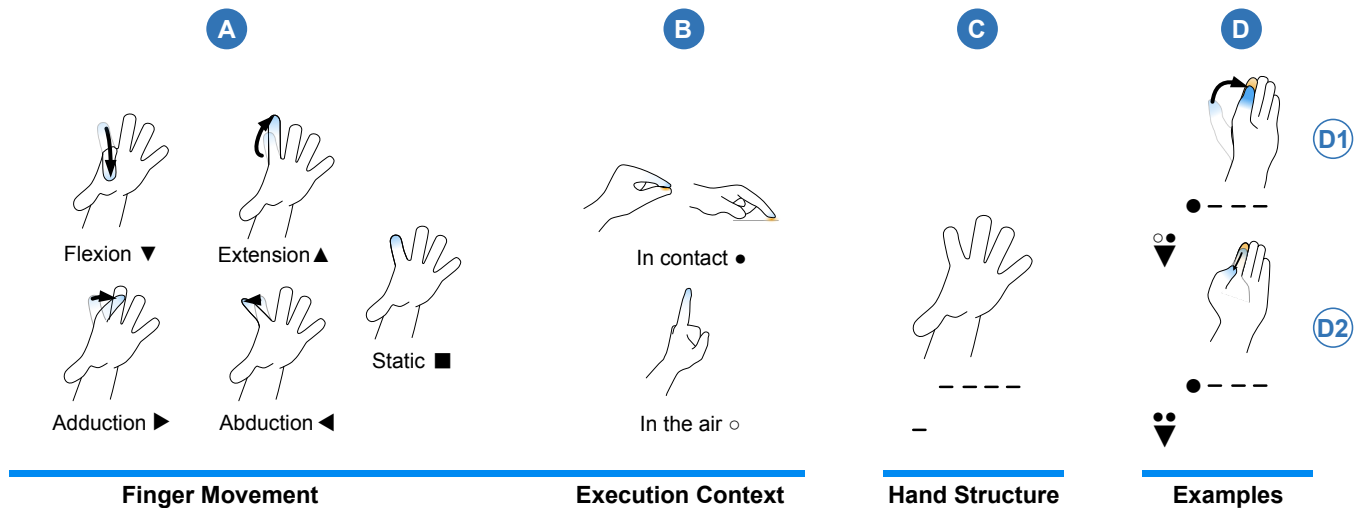


Figure 1: The μGlyph notation. Core elements: A) finger movements and B) execution contexts. C) Graphical structure based on a direct mapping with the hand. D) Two examples: D1) touch of the thumb on the index: a thumb flexion beginning in the air and ending in contact with the index, D2) downward drag of the thumb along the index: a thumb flexion beginning and ending at the contact of the index finger.

ABSTRACT

In the active field of hand microgestures, microgesture descriptions are typically expressed informally and are accompanied by images, leading to ambiguities and contradictions. An important step in moving the field forward is a rigorous basis for precisely describing, comparing, and analyzing microgestures. Towards this goal, we propose μGlyph, a hybrid notation based on a vocabulary of events inspired by finger biomechanics. First, we investigate the expressiveness of μGlyph by building a database of 118 microgestures extracted from the literature. Second, we experimentally explore the usability of μGlyph. Participants correctly read and wrote μGlyph descriptions 90% of the time, as compared to 46% for conventional descriptions. Third we present tools that promote μGlyph usage, including a visual editor with $\LaTeX$ export. We finally describe how μGlyph can guide research on designing, developing, and evaluating

microgesture interaction. Results demonstrate the strong potential of μGlyph to establish a common ground for microgesture research.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Microgesture, Notation

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1 INTRODUCTION

Over the past decade, finger movements have been explored for interaction in a variety of contexts (e.g., augmented reality, cockpits, cars) to take advantage of their speed of execution, their potential to be performed without looking at the hand, and their immediate and permanent availability while holding or not an object. This interaction modality, initially designated as finger movement [40], is now more usually referred to as hand microgesture [7, 39, 44]. In

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this paper we consider hand microgestures as finger movements only, regardless of the forearm and wrist motion.

In the very active field of hand microgestures, a consensus has emerged to describe a microgesture as a quick and subtle movement of the fingers [7, 38], but no precise definition of a microgesture is provided. The absence of a precise definition and description of microgestures leads to ambiguities and even contradictions in the papers of this field. First, ambiguity is an inherent phenomenon in natural language text+images descriptions [37, 47], which can lead to misunderstandings and inconsistencies amongst researchers. Second the current level of informality in the descriptions of microgestures leads to contradictions. For instance, the microgesture named tap can be considered as a mouse down event in [7] and as a mouse click event in [44]. Similarly, the microgesture named press can be considered as a mouse down event in [8] and as an increase of pressure in [39]. Moreover, the levels of detail in the description of microgestures is variable from one paper to another. For instance, one microgesture (e.g. tap [38]) can be decoupled into several microgestures by considering the finger that initiates the movement or the contact surface (e.g. tap of the index finger on a specific phalanx [35]). A notation to precisely define microgestures at multiple levels of granularity is required to be able to relate studies on similar microgestures and, more generally, to establish a common ground for microgesture research.

To this end, this paper proposes μ Glyph (Figure 1), a hybrid (graphic-textual) notation of microgestures for HCI researchers and practitioners. The core elements of the notation are based on the biomechanics of the fingers to ensure the completeness of movements. A microgesture description is composed of:

- 1/ a movement (or absence of movement),
- 2/ a context of execution (i.e., movement performed in the air or on a surface) or a transition between two contexts (e.g., movement beginning in the air and ending in contact with a surface), and optionally,
- 3/ characteristics further detailing the movement (e.g., which finger performs the movement).

We investigate the expressiveness of the μ Glyph notation by showing how the notation encompasses the features of existing microgesture taxonomies and by demonstrating how the notation solves ambiguities and contradictions identified in the literature. We further provide a database of 118 microgestures extracted from the literature that includes for each microgesture its provided description as well as its μ Glyph description. Given the trade-off between expressiveness and ease of use, we also examined the ease of use of the μ Glyph notation as compared to conventional text+images description. We conducted a user study with 18 participants who learned the notation, and then read and wrote μ Glyph descriptions of microgestures. After a 9-minute course and a 10-minute training session, participants correctly interpreted μ Glyph 89% of the time and correctly wrote μ Glyph 91% of the time. In comparison, when the same participants read conventional descriptions (text+images) from the literature, they correctly interpreted the microgestures 46% of the time. To promote the usage of the μ Glyph notation, we provide support tools available online: a UI with search features

for the database, a visual editor with $\LaTeX$ export, and a proof-of-concept tool that generates an animated 3D model from μ Glyph descriptions, with SVG export of the animation keyframes. Beyond an expressive and usable notation with support tools, μ Glyph can also be used as a basis for engineering microgesture interaction, as has been done with movement/gesture notations: interaction design [23], development of recognizers and toolkits [26] and evaluation with a predictive model of performance [6].

The primary contributions described in this paper are:

- an expressive notation to describe microgestures at different levels of granularity;
- an evaluation that offers empirical insights into the usability of the notation;
- a database of 118 microgestures extracted from the literature and described using μ Glyph, along with a set of support tools to promote the use of the notation.

2 RELATED WORK

Currently, there is no notation for microgesture interaction. Previous work has described microgestures with a name and a description, which is either a picture, a text or both, as in [7, 39]. Beyond punctual representations, as commonly used to represent specific gestures in different HCI fields, e.g. [16, 43] for touch gestures, we focus on notations that define rules for representing gesture features. After a review of movement/gesture notations, we review definitions and taxonomies of microgestures to highlight the variety of microgesture features. Finally, we point out the existing terminological problems due to the lack of notation.

2.1 Movement and gesture notations

The *cognitive dimensions of notations* introduced by Green [19] and further enriched by Green and Petre [18] define principles that can inform the design of a usable notation. The dimensions provide a framework for reasoning and analyzing features in notation. However, the dimensions are not independent: for a given notation a trade-off between the dimensions is achieved depending on the intended goals of the notation. Thus, a notation does not address all the dimensions and we give the short definitions of the dimensions used in the paper. **Abstraction gradient**: types and availability of abstraction mechanisms. **Closeness of mapping**: closeness of representation to domain. **Consistency**: similar semantics are expressed in similar syntactic forms. **Diffuseness / terseness**: verbosity of language.

While various notations exist for movements and gestures, we focus on notations that describe finger movements, in particular the multi-touch notations. We invite readers to read GestureCards [23] for a more detailed review of multi-touch notations. We provide a mere summary of existing notations organized into three types: textual, graphic and hybrid combining textual and graphic descriptions.

Textual notations rely on a set of letters, numbers, operators, and rules to define gestures. Choi et al propose a textual notation for 3D hand gestures that uses numerical coding to describe a hand

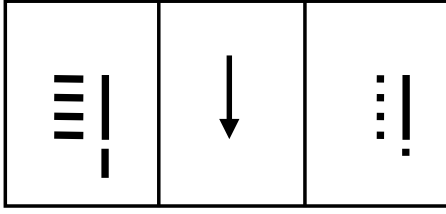


Figure 2: Example of a gesture description using the notation introduced in Charade [4]. The represented gesture is the arm moving down while closing all fingers.

pose [9]. For instance, a close hand with the index finger open is represented by 51666 ; 2422-233. It is therefore complex to read and write such a description without knowing the notation perfectly (low **closeness of mapping**). Krupka et al. also propose a XAML-based textual notation that allows developers to describe 3D hand gestures as hand pose sequences using natural language keywords such as “middle” “above” “index” [27]. However, both notations remain at finger level and it is not possible to describe a touch on a part of a finger, for instance a touch on a phalanx or on one side of a finger. Describing microgestures based on hand poses, will also lead to long and wordy descriptions of the sequence of intermediate finger poses (low **diffuseness / terseness**).

For multi-touch gestures, textual notations are often built to be computer readable in order to create gesture recognizers. To this end, Proton++ uses regular expressions and three atomic events, i.e. down (D), move (M), up (U), to define and recognize multi-touch gestures [26]. Those atomic events are represented by the structure $E_{TID}^{A1:A2:...}$ where $E \in \{D, M, U\}$, TID is the identifier of the touch and $A1:A2...$ are the properties of the touch, e.g. the trajectory direction with N (north), S (south), W (west), E (east), O (undetected direction). However, such precise textual descriptions can become complicated (low **closeness of mapping**). For instance $D_1^O M_1^O * M_1^S M_1^{OS} * M_1^E M_1^{OE} * U_1^{OE}$ describes an L-shaped trajectory. GeForMt uses a set of higher-level events defining the basic gestures, e.g. line, circle. Events can be combined to create and recognize more complex gestures [25]. For instance CONNECT_START[LINE_SW;LINE_SE] describes two fingers touching and then separating, one towards south west and the other towards south east. By using words, the descriptions are easier to read than those of Proton++ but this results in long expressions (low **diffuseness / terseness**). The use of words as in GeForMt increases the human readability of the descriptions. However, when it comes to accurately describing a gesture, using words can lead to longer and more difficult to understand descriptions than graphical descriptions.

Graphic notations rely on a set of graphic symbols that are spatially arranged. Graphic notations are often built to be human readable. In multi-touch notations, these graphical elements are often arrows, for movements (high **closeness of mapping**), and icons, for additional features. For instance, Monox uses a set of different arrows, e.g. different arrowheads, to represent basic gestures, e.g. move, swipe and drag&drop [15]. Monox is a gesture notation for collaborative design and analysis of multi-touch interactions.

Baudel et al. created Charade, an application using 3D hand gestures. To describe the gestures, they propose a notation representing the movement of the arm with the initial and final pose of the hand [4]. Graphic symbols, e.g. dots and lines, are used to indicate whether a finger is closed or not. The symbols are spatially arranged according to a representation of a hand which offers high **closeness of mapping**, see Figure 2. However, the notation is based on the description of the wrist and the pose of each finger (low **Abstraction gradient**). Describing how these fingers interact with each other, i.e. contact, would also involve introducing new symbols and redesigning the graphical structure of the notation.

Labanotation is a movement notation to describe dance and choreographies [20]. Labanotation uses a set of graphical symbols to represent the part of the body that moves, the duration, the direction, and other features of the movement. A specific subset of the notation focuses on the hand and finger movements [21]. However, it takes intensive learning and training to read and use this notation for accurate descriptions: the complete list of Labanotation graphical symbols are described in 10 different books. Therefore, it is important for a graphic notation to have a high semantic transparency, i.e. “a novice can easily infer the meaning of a symbol from its appearance” [33], to shift the cognitive load from working memory to visual. Nonetheless, Labanotation summarizes the requirements for finger movement notations: “[the notation] defines [...] different [hand] parts as well as how these flex and extend or contact objects, another hand or indeed, itself” [21]. Benesh is another dance notation [5], which represents the body as a stave (high **closeness of mapping**). For movements, only the actuators (body and limb positions) are described within the five-line stave.

Hybrid notations use textual and graphic elements to describe each part of the movement with the most appropriate representations. Graphic elements can be used to abstract concepts that require complex and/or lengthy textual descriptions [34]. Moreover, graphic symbols can depict the perceptual content of the represented concept, i.e. a high **closeness of mapping**. For instance, GestureCards [23] graphically represents the movement (high **closeness of mapping**) and uses additional textual elements to specify features including duration, direction, pressure, and speed (high **consistency**). Figure 3 presents a gesture card for a swipe left gesture.

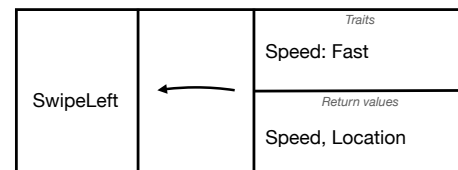


Figure 3: Example of a gesture card, reproduced from [23].

Drawing on existing notations, a hybrid notation of microgestures is promising: the notation should combine in a **consistent** way textual elements with graphic elements for a high **closeness of mapping**. Furthermore, the notation must enable the description of all possible finger movements and contacting surfaces, thus

fulfilling the requirements defined for Labanotation. The following section highlights the variety of movements and contacting surfaces that need to be described by a microgesture notation.

2.2 Variety of microgestures

Soliman et al. define a design space for thumb-to-finger and finger-to-thumb microgestures [39]. This design space is organized along four axes: the type of actions (i.e., tap, slide and draw), the finger that initiates the action, the area of the finger on which the action is performed (e.g., a phalanx or the side of the finger) and the extension state of the other fingers [39]. For example, considering these four feature axes, we can differentiate a thumb tap on the bottom phalanx of the index finger from a thumb tap on the tip of the index finger. Moreover, these two microgestures can be distinguished according to the state of the other fingers of the hand, for example if the ring finger and/or the little finger are opened and/or closed during the movement. Those four axes, namely action, actuator, surface, and other properties, can be extended to all types of microgestures and give us a structure to review the features of microgestures.

Several studies on microgestures define additional actions to those defined by Soliman et al [39]. Way et al. define thumb-to-finger touch, finger curl, finger extension, and two-finger opening/closure. These actions were defined for microgestures that are recognizable by a wrist-worn camera [11]. Chan et al. define tap, swipe, draw and circle. These actions were identified from a user elicitation study with 16 participants [7]. Other actions have been studied but never defined: for instance ream [47], tab [47], stretch [38], press [38], up [14] or flick [12].

For actuators which perform the actions, while Soliman et al. only consider a single finger [39], other studies focus on microgestures performed by a group of fingers, e.g. tap thumb with index and middle fingers [7], or go into more detail by considering a specific joint of a finger, e.g. finger lift [11].

A microgesture can be further characterized according to the surface touched by the actuator. The literature on microgestures is divided into two families: free-hand microgesture and grasping microgesture. Free-hand microgestures are microgestures performed without holding an object. In this context, the touched surfaces are often parts of the hand itself. Soliman et al. define as a surface, a finger, a finger phalanx and a finger side [39]. Other studies introduce as a surface, a group of fingers, e.g. swipe on the index and middle fingers with thumb [7], and the palm, e.g. tap palm with the index, middle and ring fingers [7]. Grasping microgestures are microgestures performed while holding an object. In this context of holding an object, Sharma et al. define three surfaces: on body, on object, in the air [38]. On body, surfaces are similar to the ones defined for free-hand microgestures, e.g. a tab where the thumb goes around a driving wheel to touch a specific finger [47]. On object, surfaces can be any surface other than the hand, e.g. external touch [11], or specific object, e.g. up / down drag of the index or middle finger on stylus [47]. In the air, microgestures are performed without touching any surface, e.g. stretch [38] and triangle [14].

Finally, other features than the extension state of the fingers [39] have been introduced. For instance, the pressure is used to distinguish between a tap and a force tap [45].

2.3 Imprecise descriptions of microgestures

The variety of features of the microgestures already studied combined with the lack of precise descriptions of microgestures, leads to misunderstanding, ambiguities and even contradictions. A microgesture is generally described by a name and a description in the form of natural language text and/or images.

First, while some papers include a detailed textual description of each microgesture [7, 11, 39, 45], many papers describe a microgesture only by a name and an image [8, 12, 14, 24, 30, 31, 37, 38, 47]. This lack of precision can lead to misinterpreting a microgesture, for instance the microgesture named ream [47]. Furthermore, the same image can be used for describing different microgestures [47] leading to even more ambiguity.

Second, the description often lacks a precise definition of the initial and final states of a microgesture. For instance, does a tap [7, 44] end in contact or in the air? This issue with initial and final states consists of identifying which part of the movement is key to describing the microgesture. We illustrate this issue by making the analogy with a mouse down event and a click event which consists of a mouse down then up. Is the mouse down the key part of the movement or is it both mouse down and mouse up that define the key part of the movement? For instance, a touch [47], a tab [47], a tap [7] and a press [8] can be considered as a mouse down event while a tap [44] as a click event.

The absence of precise description leads to naming issues. Such issues are serious because they make it difficult to relate and compare pieces of research. Indeed, a same microgesture can be designated by several names, as illustrated by the following three examples (precisely described in Table 7):

- close gap of index and middle [11], tap of index and middle [7]
- drag[47] and slide [39]
- multi finger snap [12] and swipe across all fingers [7]

Moreover, the same name can designate different microgestures. In the following three examples, we quote names used for different microgestures and we provide their precise descriptions that highlight their differences in Table 7:

- extend [11] and extension [37]
- tap [7] and tap [44]
- press [8] and press [38]

The need for precise descriptions based on a notation is thus paramount in moving the field forward. Given the variety of features of the microgestures already studied, we conclude that a microgesture notation must:

- enable the unambiguous description of a large variety of microgestures, not covered by existing taxonomies;
- enable the description of microgestures at several levels of abstraction (**Abstraction gradient** dimension [18]), for example at a high level, by ignoring which finger is touched

		Choi et al. [9]	Krupka et al. [27]	Proton++ [26]	GeForMt [25]	Monox [15]	Charade [4]	Labanotation [21]	GestureCards [23]	Text/Images [7, 39]	μGlyph
Cognitive dimensions	Abstraction gradient description at different levels of abstraction	-	-	-	-	-	-	~	~	+	+
	Consistency specifically, robust to new features	-	~	~	~	+	-	+	+	-	+
	Closeness of mapping clear mapping between representation and movement	~	~	-	+	+	+	-	+	+	+
	Diffuseness / terseness concise description of a movement	+	-	+	-	-	-	+	-	-	+
Movement features	Description of the initial and final states of the movement	+	+	+	~	~	+	+	-	~	+
	Description of the fingers	+	+	-	-	-	+	+	-	+	+
	Description of other subparts of the hand (phalanxes, finger sides, palm, ...)	-	-	-	-	-	-	+	-	+	+
	Features of the movement other than its direction (pressure, amplitude, duration, ...)	-	-	-	-	~	~	~	+	+	+

Table 1: Desired cognitive dimensions and movement features listed as requirements for a microgesture notation. Requirements and existing notations: ‘+’: the notation fulfills the requirement, ‘~’: the notation partially fulfills the requirement, ‘-’: the notation does not fulfill the requirement.

by another finger and at a low level by describing the finger that performs the action, the finger that is touched or other features such as the exerted pressure or the extension state of a finger;

- have a stable set of core elements that can be consistently (**Consistency** dimension [18]) extended with new features as the research area progresses (e.g., nail regions as new contacting surfaces [30]);
- be easy to read and write to be widely adopted by the community (**Closeness of mapping** and **Diffuseness / terseness** dimensions [18]).

Based on these requirements for a notation, Table 1 summarizes our review of existing notations. The following section describes μGlyph, the new notation we propose to fill the gaps by meeting all the requirements.

3 μGLYPH NOTATION

μGlyph is a hybrid notation describing all possible finger movements and contacting surfaces. Referring to the *cognitive dimensions of notations* [18], the design of the μGlyph notation is as follows. μGlyph is based on a graphical representation of the hand (high **closeness of mapping**) where only the key elements of the microgesture are represented. To this end, the graphical symbols correspond to a finite set of low-level events that concisely (high **diffuseness / terseness**) describe any type of movements. A microgesture is a sequence of finger movements within an execution context. Graphical symbols represent the execution context that specifies if

Heavy touch	Down drag	Up drag	Release
			
⬇	⬇	⬆	⬆

Table 2: Examples of microgestures described at a high level of abstraction with the movement and context glyphs of μGlyph.

the movement is done in the air or in contact with a surface. The textual description is used in a consistent way (**consistency**) for finer features of the movement (**abstraction gradient**). The textual description also contributes to the extensibility of the notation, making it easier to add new movement features in the future.

Having introduced the main design principles of the μGlyph notation, we present it now step by step. Figure 4 at the end of the section summarizes all the steps.

3.1 Movements, Contexts and Graphical Structure

Based on the biomechanics of a finger, we use the finger movements described in Figure 1. A glyph is associated with each of these movements, derived from anatomical observations [42]. The

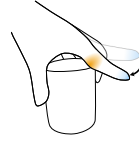
Hand structure	Index flexion	MCP (metacarpophalangeal) index flexion	PIP (proximal interphalangeal) index flexion	Base index touch
				
-----	oo-----	oo----- MCP	oo----- PIP	oo----- b

Table 3: Examples of microgestures described at a low level of abstraction by specifying the features of the actuator as defined in μ Glyph.

following five movement glyphs are core elements of the μ Glyph notation:

- Extension (▲):** movement of a finger away from the palm;
- Flexion (▼):** movement of a finger closer to the palm;
- Abduction (◄):** movement of a finger away from the sagittal plane;
- Adduction (►):** movement of a finger closer to the sagittal plane;
- Static (■):** absence of movement.

Movement glyphs are inspired from the representations commonly used for the concepts *play* (► movement metaphor) and *stop* (■ no movement metaphor). The triangle orientation represents the direction of the movement.

These finger movements are performed in an execution context that characterizes whether the movement is performed in the air or in contact with a surface. To represent the contact with a surface, we chose the glyph ● (full circle), as in many representations of multi-touch gestures, e.g. guide iOS, wikipedia "Pointing device gesture". Contrastingly, a finger in the air is represented with the glyph ○ (empty circle). We describe the contexts only at the beginning and at the end of the movement, which represents six possible cases divided into two groups.

(1) Single-glyph contexts

- : the finger is in the air and stays in the air, in the same position. There is no movement possible.
- : the finger stays at the same position on a surface. Some movements are possible, all subtle. For example, to exert pressure on a surface, the finger must be moved in the direction of the surface, but the contact position remains unchanged.

(2) Dual-glyph contexts

- : the finger moves from one position in the air to another.
- : the finger moves from a position in the air to a position in contact with a surface.
- : the finger moves from a position in contact with a surface to a position in the air.
- : the finger moves from one position in contact with a surface to another.

Extension (▲), Flexion (▼), Abduction (◄) and Adduction (►) are used with the contexts for which a movement is possible (○○,

○○, ●○, ●● or ●●). Static (■) is used only with the two contexts without changing position (○ or ●). Context glyphs are placed above a movement glyph. Two graphical structures co-exist: — for single-glyph contexts and — for dual-glyph contexts. The glyph — (dash) represents an unspecified slot, a wildcard for any glyph.

As letters form words, groups of glyphs form events. An event is a combination of movement and context glyphs (e.g. ▼●). Such combinations of glyphs allow microgestures to be described at a high level of abstraction, as illustrated in Table 2 with elementary microgestures consisting of one event. By using unspecified slots for movements and/or contexts, microgestures can be described at an even higher level of abstraction. For instance, a drag can be described as the movement of a finger from one position in contact with a surface to another, without specifying the direction. The corresponding μ Glyph description is: —●. This can be a drag up ▲ or a drag down ▼, for example.

To obtain a more detailed description of a microgesture (low level of abstraction), μ Glyph includes movement features, such as the actuator (the part of the hand that performs the movement), and context features, such as the receiver (part of the hand on which the movement is performed). Below we present these features of events and explain how to combine events to describe microgestures composed of several events.

3.2 Movement features: actuator

The actuator represents the part of the hand, i.e. the finger, which performs the movement [7, 11, 24, 36, 39, 45, 47]. To specify the finger actuator, we adopt a graphical structure representing a hand with the positions of the fingers: ----- (from left to right: thumb, index, middle, ring, pinky; see the representation on the left in Table 3). Just like the graphical structure of an event, a hand-based description can include unspecified slots (—). When specified, a slot defines an event. For instance, a flexion of the index finger which results in contact with a surface is described as follows: —▼●—.

For more detailed descriptions, other features of the actuator are specified as indices of the event. The parts of the finger that enter in contact with a surface are represented by the following textual symbols: b (*base*), m (*middle*), t (*tip*) and n (*nail*) as indices of the event. For instance, a touch performed by the base phalanx of an

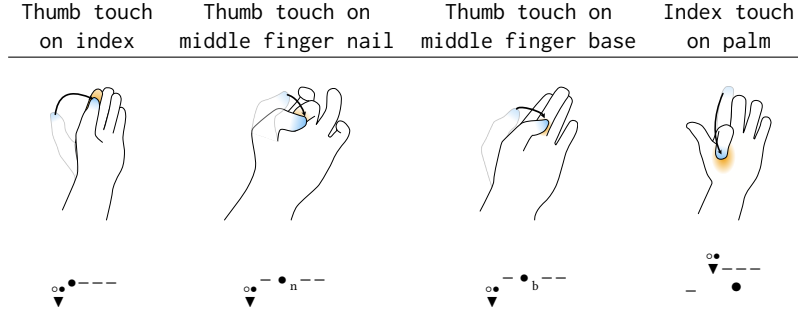


Table 4: Examples of microgestures described at a low level of abstraction by specifying the features of the receiver as defined in μGlyph.

index finger is written as $\overset{\circ}{\underset{\circ}{\nabla}}_b---$. Additional symbols denote the sides of a finger performing the movement: *l* (left), *r* (right), *f* (front) and *d* (dorsal). For instance, a touch performed by the left side of an index finger is written as $\overset{\circ}{\underset{\circ}{\nabla}}_l---$. Phalanges and sides can be combined, for instance a touch performed by the left side middle phalanx of an index finger is written as $\overset{\circ}{\underset{\circ}{\nabla}}_{m,l}---$. In addition to the finger parts, the finger joints performing the movement can be specified. For example, the *lift* microgesture, presented in EMGRIE [11], is performed by a specific finger joint. To describe the finger joint, we introduce two indices MCP (*Metacarpophalangeal*) used in EMGRIE, and PIP (*Proximal interphalangeal*) never used in the microgesture literature yet. MCP is the joint between the finger and the palm. PIP is the joint between the finger phalanges. The *lift* microgesture [11] is written as $\overset{\circ}{\underset{\circ}{\nabla}}_{MCP}$.

Table 3 shows some of the examples presented above, including for each microgesture, its graphical representation and its μGlyph description.

3.3 Other movement features

Beyond the actuator of the movement, we introduce features of the movement itself.

First, when in contact with a surface, the actuator can exert pressure on the surface [11, 38, 44, 45, 48]. While in theory we can detect an infinite number of pressure levels, in practice, and without visual feedback, a user can only distinguish between a limited number [46]. In the microgesture literature, there are at most three pressure levels [45]. We represent those three pressure levels by Lo, Md and Hi, respectively for a *low*, *medium*, or *high* level of pressure. If only two levels are used, we then recommend using Lo and Hi. In the μGlyph notation, the level of pressure is specified below the movement glyph. For instance, a constant high pressure drag on a surface is written as $\overset{\circ}{\underset{\circ}{\nabla}}_{Hi}$. If the level of pressure changes between the beginning and the end of the movement, the change in pressure is specified with an arrow between the two pressure levels. For instance, the increase in pressure of a finger already in contact with a surface is written as $\overset{\circ}{\underset{\circ}{\nabla}}_{Lo \rightarrow Hi}$. If required, new

pressure levels can be defined and pressure values can be specified, e.g. an increase of pressure from 1 to 3 *Newton* $\overset{\circ}{\underset{\circ}{\nabla}}_{1N \rightarrow 3N}$.

Second, microgestures are described as quick and subtle movements [7, 38, 39], corresponding to two additional movement features: duration and amplitude of the movement. Even though these two features are rarely used in the microgesture literature, they can be specified below the movement glyph. For instance, a maximum dwell time of 40 ms without movement is written as $\overset{\circ}{\underset{\circ}{\nabla}}_{t < 40ms}$.

amplitude can be used to specify a movement. From our literature review, three amplitudes of movement are identified: Cl (*Close*), Ha (*Half*) and Op (*Open*). For instance, a finger extension ending with a half-open finger is written as $\overset{\circ}{\underset{\circ}{\nabla}}_{Op \rightarrow Ha}$ and a flexion from a fully open

finger to a half-open finger as $\overset{\circ}{\underset{\circ}{\nabla}}_{Op \rightarrow Ha}$. In addition, a finger pose (for example a half-open finger) is written as $\overset{\circ}{\underset{\circ}{\nabla}}_{Ha}$. It is also possible to specify the opening percentage of the finger. A finger extension ending to a 25% open finger is written as $\overset{\circ}{\underset{\circ}{\nabla}}_{\rightarrow 25\%}$.

3.4 Context features: receiver

The receiver is the contact surface. The μGlyph features of a receiver are therefore only used in contexts where a contact surface is involved. Based on the literature, μGlyph enables us to describe two types of contact surfaces: a part of the hand, e.g. [39] and an object, e.g. [38].

When the contact surface is a part of the hand, it is described by the contact glyph $\bullet$ within the graphical representation of the hand. For instance, the thumb touching the index finger is written as follows: $\overset{\circ}{\underset{\circ}{\nabla}}_{\bullet}$. For more detailed descriptions, the features used for the actuator apply to the receiver. The textual symbols *b* (*base*), *m* (*middle*), *t* (*tip*) and *n* (*nail*) are therefore also used to describe the receiver. For instance, the thumb touching the base of the middle finger is written as $\overset{\circ}{\underset{\circ}{\nabla}}_{\bullet b}$ and the thumb touching

represent a monotonic progression of other movement features (for instance $\overset{\circ}{\blacktriangledown} ; \overset{\circ}{\blacktriangleright} ; \overset{\circ}{\blacktriangle} ; \overset{\circ}{\blacktriangleright}$ encapsulated into $\overset{\circ}{\square}$).
1N → 2N 2N → 2N 2N → 3N 3N → 4N 1N → 4N

3.7 Concise notation

So far, we have described a microgesture using a graphical representation of the hand to make the description more intelligible. However, this notation can take a lot of space to describe a simple microgesture, such as an extension of the index finger: $\overset{\circ}{\blacktriangle}---$. For a more concise and advanced way of describing a microgesture, the fingers are specified by their first letter at the bottom left of the event glyph. For instance, an extension of the index finger is written as $\overset{\circ}{\blacktriangle}_i$. Receivers can also be specified in a more concise way. They are represented as a function of the event. For instance, a touch of the middle finger on the base phalanx of the thumb is written as $\overset{\circ}{\blacktriangledown}_m(\bullet_b)$. The rest of the notation is unchanged.

Table 6 includes microgestures and their two descriptions: the hand-based description and the concise description.

4 THEORETICAL EXPRESSIVENESS: DESCRIPTIVE POWER OF μGLYPH

Theoretical expressiveness refers to the ability of μGlyph to unambiguously describe existing microgestures, regardless of the usability of μGlyph. Indeed, the ease of use, reading and writing of μGlyph descriptions, i.e. the practical expressiveness of μGlyph, is the focus of the next section.

First, we motivate the μGlyph notation by showing that it encompasses and extends the axes of Soliman et al.'s characterization scheme [39]. The four axes, namely *Action*, *Actuator*, *Touch Location* and *Finger flexion*, are described in the related work section. The first three axes correspond in μGlyph respectively to *the movements and contexts*, *the movement features: actuator* and *the context features: receiver*. In addition, the μGlyph *movement features: actuator* extend Soliman et al.'s *Actuator* axis by considering the parts of the finger (phalanxes, sides) and the finger joints. The fourth axis, *Finger flexion*, describes the extension state of the fingers that are not the actuator and corresponds to the μGlyph *other movement features*. As presented by Soliman et al., there are two ways to describe a *finger flexion*, by specifying either the flexion movement or the finger pose. Using μGlyph, on the one hand, a flexion movement is described using Extension ($\blacktriangle$) and Flexion ($\blacktriangledown$), for instance a finger extension ending with a x% open finger $\overset{\circ}{\blacktriangle} \rightarrow x\%$. On the other

hand, a finger pose is described with Static ($\blacksquare$), for instance a x% open finger $\overset{\circ}{\blacksquare}_{x\%}$. More generally, this example shows that μGlyph supports the two common ways of describing a microgesture: the description of a sequence of movements or a sequence of positions.

Second, we translated into μGlyph all the microgestures described in a precise and textual way in the literature [7, 11, 39, 45]. The purpose of this translation exercise was to test the ability of the notation to allow the specification of subtle differences described in the text. Table 7 includes a subset of these microgestures. Using

μGlyph, we described several other microgestures of the literature. The tables in the Appendix B contain the complete list of microgestures described in μGlyph. In total, we described 118 microgestures from 17 papers. Of the papers considered, some provide precise definitions and characterization schemes of microgestures [7, 11, 39, 45], others are seminal papers in the field of microgesture e.g. [38, 47]. In addition, the papers cover various contexts of use, e.g. airplanes [44], cars [14, 22, 47], microgestures on a pen [40], free-hand microgestures [8, 12, 24] and propose various recognition mechanisms, e.g. cameras [8], gloves [44], bioacoustics [12] and electromyography [36]. Some papers also provided us edge cases to describe using μGlyph, e.g. [30].

Finally, we have pointed out some terminological problems existing in the literature. It is worth examining whether the μGlyph notation, by allowing a precise and unambiguous description of the microgestures, solves these problems. Based on the description in μGlyph, we can identify similar microgestures that are referred to by different names in the literature. For instance, a close gap index and middle [11] and a tap of index and middle [7] are both described by $\left(\overset{\circ}{\blacktriangleright}_i(\bullet_m) \quad \overset{\circ}{\blacktriangledown}_m(\bullet_i) \right)$. In addition, a μGlyph description can help to distinguish microgestures designated with the same name: extend [11] $\overset{\circ}{\blacktriangle}_{MCP}$ and extension [37] $\overset{\circ}{\blacktriangle}(\bullet)$ or Tap [7] $\overset{\circ}{\blacktriangledown}$ and Tap [44] $\overset{\circ}{\blacktriangleright}$; $\overset{\circ}{\blacktriangle}$. Table 7 presents other examples of ambiguity.

To facilitate the work of researchers and practitioners, we have created a database compiling each of the microgestures presented in the tables in Appendix B. For each microgesture, the database contains the following information: name used in the paper, textual description, visual description, paper title and DOI, μGlyph description, intended context of use (i.e. whether the microgesture is performed free-hand or while holding an object). In addition, if the microgesture is performed while holding an object, information on the grasps and objects studied is included.

5 PRACTICAL EXPRESSIVENESS: USABILITY OF μGLYPH

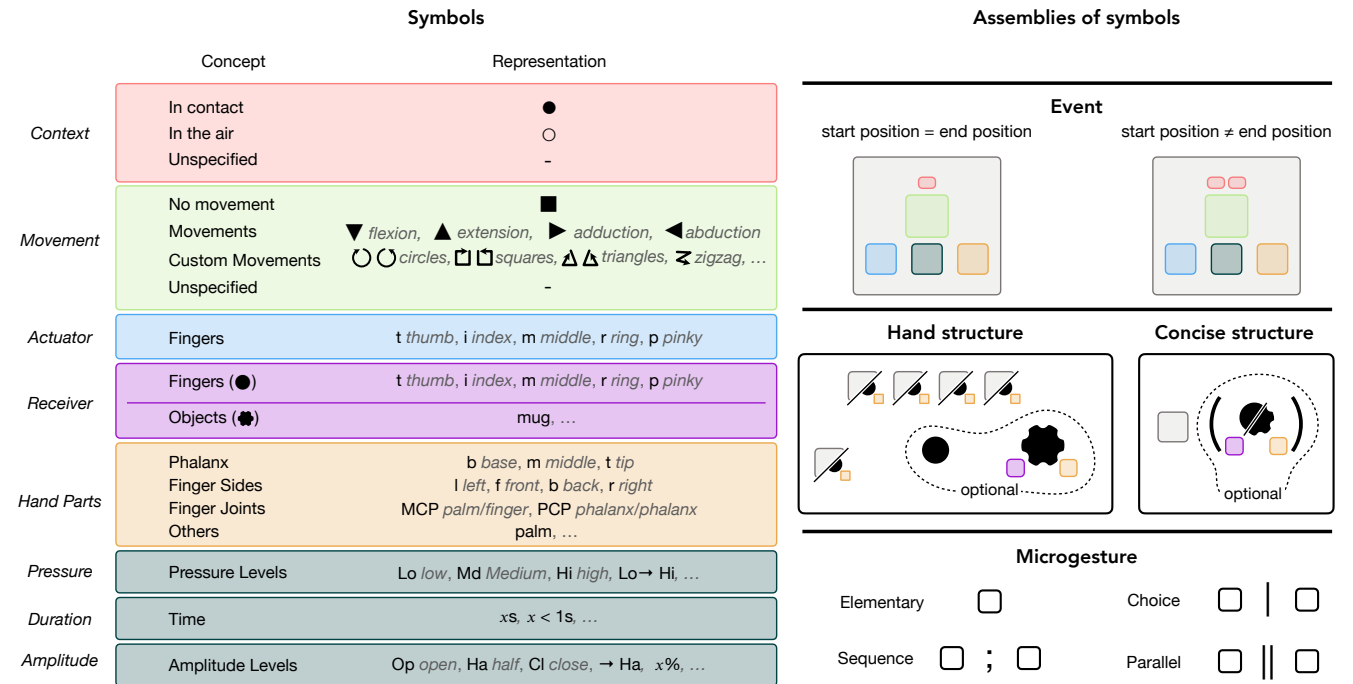
The practical expressiveness refers to the usability of the μGlyph notation by researchers and practitioners. As identified for task modeling notations in [3], the usability of a notation covers two distinct notions. The first notion refers to the ease with which a μGlyph description can be read and understood. This notion of usability integrates that of learnability and readability. The second notion concerns the ease with which μGlyph descriptions are generated and modified. We structure the experimental study according to these two notions of usability: reading and writing. For the reading part, the experiment is inspired by the first experiment conducted by Heseniuss et al. to evaluate the readability of GestureCards, a gesture notation [23].

5.1 Participants

The end-users of μGlyph are HCI researchers and practitioners. Therefore, we decided to recruit participants who work in HCI related fields and have an academic background. We recruited 18

Table 6: Microgestures described using the standard μ Glyph hand-based representation and using the concise μ Glyph representation.

Microgesture name	Standard μ Glyph	Concise μ Glyph
Index base touch		
Thumb touch on the nail of the middle finger		
Index touch on the palm		
Index base touch on a cup		
Left swipe of the thumb on the index base or middle phalanx	$x \in \{b m\}, \text{---} \bullet \text{---} ; \text{---} \bullet \text{---} ; \text{---} \bullet \text{---}$	$x \in \{b m\}, \text{---} \bullet \text{---} ; \text{---} \bullet \text{---} ; \text{---} \bullet \text{---}$
Index and middle touch on the thumb		
Square made by the index finger in the air		

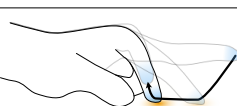
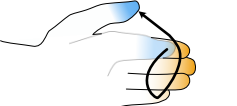
**Figure 4: Global summary of the μ Glyph notation.**

unpaid participants (30 y.o $\pm$ 9): 6 HCI PhDs, 4 HCI PhD students, 4 HCI internship students, 2 HCI research engineers and 2 PhD students from another domain who work with notations and have a strong interest in HCI. For context, 6 participants had had a course with an exam less than a year ago, 4 less than 5 years ago, and 8 more than 5 years ago (7 years ago $\pm$ 8).

5.2 Setup

Participants sat at a desk in front of a computer. Descriptions and videos were displayed on the computer screen. Descriptions and videos used in the experiment are available in the supplementary material of the paper. A camera was placed in front of the participants and only captured their hands. Participants used the keyboard for pressing the RETURN and SPACE keys. Pen and paper were placed on the desk for participants to write their answers.

Table 7: Theoretical expressiveness of μGlyph: descriptions of microgestures at two levels of abstraction and highlighting (in bold) of terminological problems existing in the literature.

Picture	Paper	μGlyph	Additional characteristics
	Press [8] Tap [7] Finger touches [11] Finger taps [39]	$\overset{\circ}{\underset{\circ}{\nabla}}(\bullet)$ $\overset{\circ}{\underset{\circ}{\nabla}}$ $(x, y) \in A, \overset{\circ}{\underset{\circ}{\nabla}}(x, y)$	$A = \{i m r p\} \times \{b m f\}$ $A = \{i m r\} \times \{b m f\}$
	Tap [38] Tap [37] Tap [44] Pinch [36]	$\overset{\circ}{\underset{\circ}{\nabla}}; \overset{\circ}{\underset{\circ}{\blacktriangle}}$ $(x, y) \in A, \overset{\circ}{\underset{\circ}{\nabla}}(x, y); \overset{\circ}{\underset{\circ}{\blacktriangle}}$	$A = \{i m\} \times \{b m f\} \cup \{i, tl\} \cup \{r, n\}$ $A = \{i m r p\}$
	Swipe Left or Right [37] Extension [37] Drag [47] Slide [39] Slide thumb on finger [11] linear thumb-to-finger slides [39]	$\overset{\circ}{\underset{\circ}{\blacktriangle}}; \overset{\circ}{\underset{\circ}{\blacktriangle}}$ $\overset{\circ}{\underset{\circ}{\blacktriangle}}$ $\overset{\circ}{\underset{\circ}{\nabla}}; \overset{\circ}{\underset{\circ}{\blacktriangle}}; \overset{\circ}{\underset{\circ}{\blacktriangle}}; \overset{\circ}{\underset{\circ}{\blacktriangle}}$ $x \in A, \overset{\circ}{\underset{\circ}{\nabla}}(x, \bullet); \overset{\circ}{\underset{\circ}{\blacktriangle}}(x, \bullet)$	$A = \{i m r p\}^n$, with $n \geq 1$ $A = \{i m r p\}$
	Swipe [7] Drag&Drop thumb on index-side [47]	$x \in A, \overset{\circ}{\underset{\circ}{\nabla}}(x, \bullet); \left(\overset{\circ}{\underset{\circ}{\nabla}}(x, \bullet); \overset{\circ}{\underset{\circ}{\blacktriangle}}(x, \bullet) \right); \overset{\circ}{\underset{\circ}{\nabla}}$	$A = \{i (i m) (m r)\}$ $A = i-l$
	Swipe across all fingers with thumb [7] Multi-finger snap [12]	$\overset{\circ}{\underset{\circ}{\nabla}}(p, \bullet); \overset{\circ}{\underset{\circ}{\blacktriangle}}(r, \bullet); \overset{\circ}{\underset{\circ}{\blacktriangle}}(m, \bullet); \overset{\circ}{\underset{\circ}{\blacktriangle}}(i, \bullet); \overset{\circ}{\underset{\circ}{\nabla}}$	
	Squeeze/Pull up [36] Press [38]	$x \in A, \overset{\circ}{\underset{\circ}{\nabla}}_{Lo \rightarrow Hi}(\bullet); \overset{\circ}{\underset{\circ}{\blacktriangle}}_{Hi \rightarrow Lo}$	$A = \{i m r p\}$
	Strech [38] Extend [11] Up [14] Thumb up [47]	$\overset{\circ}{\underset{\circ}{\blacktriangle}}_{MCP}$ $x \in A, \overset{\circ}{\underset{\circ}{\blacktriangle}}_x$	$A = \{i m (r p)\}$ $A = t$

5.3 Sets of microgestures for comparison

To compare different ways of describing microgestures, we first defined 4 sets of 10 microgestures from the literature. 3 sets are used for the reading part (sets A, B and C) and 1 set for the writing part (set W) of the experimental study. The sets were designed to reflect the most common microgestures in the literature. Each set contains (listed below in this order): 3 taps/touches, 3 swipes/slides/drags, 1 pressure- or amplitude-based microgesture, 2 ambiguous microgestures, and 1 drawing microgesture. We distinguish two types of ambiguity. First, a microgesture whose name may induce a misinterpretation, e.g. Extension (Set C 8). Second, a microgesture

whose name is in conflict with another microgesture of the same set, e.g. Robin presses (Set A 8), in μGlyph $\overset{\circ}{\underset{\circ}{\nabla}}(i, \bullet)$, and Press (Set A 7), in μGlyph $\overset{\circ}{\underset{\circ}{\nabla}}_{Lo \rightarrow Hi}$. Moreover, each set uses similar features or combination of features while covering a broad range of μGlyph symbols. For instance, the first microgesture of each set is a Tap thumb with finger “x” (single actuator), with finger “x” different for each set. The second microgesture is a Tap with different parallel actuators. Finally, for the reading part of the study (sets A, B and C), each set contains 5 microgestures described with text and images in the original papers and 5 with images only. The

5 microgestures with images only are: 5, 7-10 in sets A and B, and 3-4, 7-9 in set C.

Set A:

- 1: Tap thumb with ring finger [7]
- 2: Tap thumb with index and middle fingers [7]
- 3: Finger Tap [39]
- 4: Swipe accross all fingers with thumb [7]
- 5: Drag and Drop Index On Thumb [47]
- 6: Rotational thumb-to-finger slide [39]
- 7: Press [38]
- 8: Robin presses on the tip of his index finger [8]
- 9: Tab [47]
- 10: Triangle CW [14]

Set B:

- 1: Tap thumb with index finger [7]
- 2: Tap thumb with ring and pinky fingers [7]
- 3: Fist Tap [39]
- 4: Swipe middle and ring fingers with thumb [7]
- 5: Drag thumb around object [47]
- 6: Linear thumb-to-finger slides [39]
- 7: Thumb up [47]
- 8: Swipe left [37]
- 9: Multi finger snap [12]
- 10: ZigZag [37]

Set C:

- 1: Tap thumb with middle finger [7]
- 2: Tap all fingers together [7]
- 3: Snip [47]
- 4: Swipe right [37]
- 5: Drag and Drop middle on index [47]
- 6: Fingertip slide [39]
- 7: Force tap [45]
- 8: Extension [37]
- 9: Ream [47]
- 10: Draw circle [39]

Set W:

- 1: Tap thumb with pinky finger [7]
- 2: Tap palm with index, middle and ring fingers [7]
- 3: Tap and Flap [39]
- 4: Swipe index and middle fingers with thumb [7]
- 5: Drag Index or Middle finger on stylus up/down [47]
- 6: Linear finger-to-thumb slides [39]
- 7: Stretch [38]
- 8: Pressing down on the lateral side of the upper index finger [44]
- 9: Flexion [37]
- 10: Square [14]

5.4 Procedure

A session consisted of two phases: a training phase and an experiment phase, see Figure 5. During the training phase, we informed

the participants about the context of the research, and defined what a microgesture is. We also explained the purpose of the experiment which is to read microgesture descriptions and transcribe microgestures using μ Glyph. After participants completed a consent form, we then gave them a hard copy of a cheatsheet (Appendix A) that they would use during the experiment. Participants then watched a 9-minute video¹ explaining μ Glyph. To conclude the training phase, the participants had to describe 5 microgestures in μ Glyph. The experimenter performed each microgesture in front of the participants, as many times as necessary, and helped them find the correct answers. The microgestures were: a touch with the thumb $\overset{\circ}{\nabla}_t$, a drag up or down of the thumb on the ring finger $\overset{\circ}{\nabla}_t(\bullet)|_t\overset{\circ}{\nabla}_r(\bullet)$, a flexion of the index finger until mid-closure $\overset{\circ}{\nabla}_i$, an extension $\overset{\circ}{\nabla}_i$ followed by a touch on the thumb with either the index or the middle finger $x \in \{i|m\}, \overset{\circ}{\nabla}_x(\bullet)|_x\overset{\circ}{\nabla}_t(\bullet)$ and a squeeze of every finger on an object $\overset{\circ}{\nabla}_t(\bullet)|_t\overset{\circ}{\nabla}_p(\bullet)$. Up to this point, participants could

ask questions on μ Glyph. Afterwards, this was forbidden. The hard copy of the cheatsheet was taken back by the experimenter. The training phase lasted around 20 minutes in total.

The experimental phase was divided into two parts: reading and writing. For the reading part, participants were asked to read a microgesture description and perform the corresponding microgesture. We used three types of description: μ Glyph, concise μ Glyph and text+images from the original paper. For each description type, we used one of the sets A, B or C. We used a Latin square to counterbalance the order of the description and the set used. The microgestures of each set were presented in a randomized order. To start a trial, participants pressed the RETURN key which prompted a description on the screen. When they understood the microgesture, participants pressed RETURN again. The time spent between both presses on the RETURN key (READING TIME) was recorded. Then, the description was hidden, and participants performed the microgesture in front of a camera while the experimenter wrote down the performed microgesture using μ Glyph. We also asked them to verbally describe the microgesture, and to explicitly state when it began and ended. Finally, participants rated their level of confidence in their response on a 6-point Likert scale (1-not at all confident, 6-very confident).

For the writing part, participants were asked to describe 10 microgestures in μ Glyph. We used the microgesture set W and presented each microgesture in a randomized order. Participants watched a video for each microgesture. They could watch it as many times as they wanted. Then, they had to verbally describe the microgesture to the experimenter to ensure that any errors in the μ Glyph description were not due to the participants describing a different microgesture than the one requested. Then, participants had to describe the microgesture in μ Glyph. To do so, participants first pressed the RETURN key and then began to write the μ Glyph description on paper. We asked them to verbally describe the microgesture as they wrote. The experimenter re-explained the microgesture if the participant verbally described a different one. We have not imposed a specific form for writing in μ Glyph, i.e. standard or concise.

¹The URL is removed for submission and the video is included in the supplementary material.

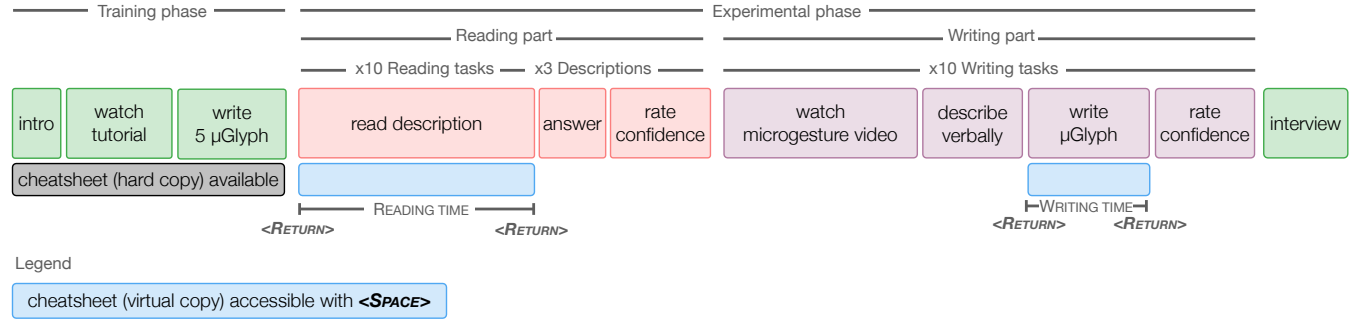


Figure 5: Experimental procedure (<RETURN> represents the return key, <SPACE> the space key).

After completing their writing, participants pressed the RETURN key again. The time spent between both presses on the RETURN key (WRITING TIME) was recorded. Then, they rated their level of confidence on a 6-point Likert scale (1-not at all confident, 6-very confident).

During both reading and writing in μGlyph, participants could freely access a virtual copy of the cheatsheet by holding down the SPACE key. The number of times the cheatsheet was used and the time spent using it were recorded. At the end of the experiment, we asked participants which form of μGlyph they preferred, why and if they had any additional comments.

Figure 5 summaries the experimental procedure. In total, the experiment lasted around 1h.

5.5 Measures and Hypotheses

During the experiment, we recorded the following measures: the number of cheatsheet calls, the total time using the cheatsheet, the total time (READING/WRITING TIME) spent on a microgesture and participants' self-confidence ratings. For the reading part, the experimenter translated the participants' answers into μGlyph using his notes, the participants' explanations, and the videos. Movement and context symbols as well as sequence and parallel combinations were inferred from the microgesture performed by the participant. The choice of events and additional features were inferred from participants' oral explanations. For instance, for a Finger Tap (Set A 3), a participant performed a tap with the thumb on the tip of the index in front of the camera, then verbally explained that this could be done on the index, middle or ring finger and on the tip, middle or base phalanx. If the translated answer had additional, missing or incorrect symbols, the answer was considered incorrect. For the writing part, we scored a μGlyph description as incorrect if it was missing one or more symbols or if the structure was wrong. If participants were more accurate than expected but their μGlyph description still correctly described the microgesture presented in the video, e.g. Tap thumb tip with pinky finger instead of Tap thumb with pinky finger (Set W 1), the answer was considered correct. Using these measures, our first hypothesis is that μGlyph performs better than text+images descriptions in terms of answer correctness and that participants are more confident in their responses. Our second hypothesis is that μGlyph performs worse than text+images descriptions in terms of how long it takes

to read a description. Our final hypothesis is that μGlyph can be used easily after a short training period, i.e., a high percentage of correct answers, and a limited number of calls to the cheat sheet and time spent on it.

6 RESULTS

The data analysis follows recommendations for "fair statistical communication in HCI" [13]. Reported confidence intervals are 95% confidence intervals computed with the formula $CI = 1.96 * \frac{\sigma}{\sqrt{n}}$, with σ the standard deviation and n the size of the considered sample. We also report the proportion overlap (POL) [10] when comparing two CIs, i.e. overlap amount / mean size of the CI arms that overlap. Using the TIP 24 of the guidelines [13], if $POL < 0.25$, i.e. if the CIs visually overlap by more than 1/4 of their length, then a reasoning similar to that of a p-value less than 0.05 can be used. Figure 6 presents an overview of the results.

6.1 Reading

Time: Overall, reading text+images descriptions of microgestures was the fastest ($24s \pm 3$, $n = 60$), followed by standard μGlyph ($36s \pm 4$, $n = 60$) and concise μGlyph ($42s \pm 5$, $n = 60$). The difference between text+images and standard μGlyph (POL: -1.4) as well as between text+images and concise μGlyph (POL: -2.5) is significant, but the difference between standard and concise μGlyph (POL: 0.66) is not.

Correctness: For correctness, CIs are computed on the success rates of participants. Reading standard μGlyph led to the highest number of correct answers ($90\% \pm 3$, $n = 60$), followed by concise μGlyph ($88\% \pm 3$, $n = 60$) and text+images ($46\% \pm 4$, $n = 60$). There is a significant difference between text+images and μGlyph (POL: -10.9) as well as between text+images and concise μGlyph (POL: -11.1), but not between standard and concise μGlyph (POL: 1.4). There was no impact of the microgesture set on correctness (POL: -1.15 when comparing two sets for the same type of description).

Considering both μGlyph forms together, there are a total of 39 errors out of 360 readings of μGlyph descriptions. We observed the following error types:

- Choice symbol "[" mistaken for sequence symbol ";" (9 / 39)
- A symbol not taken into account (8 / 39)
- Parallel symbol "||" mistaken for choice symbol "[" (5 / 39)

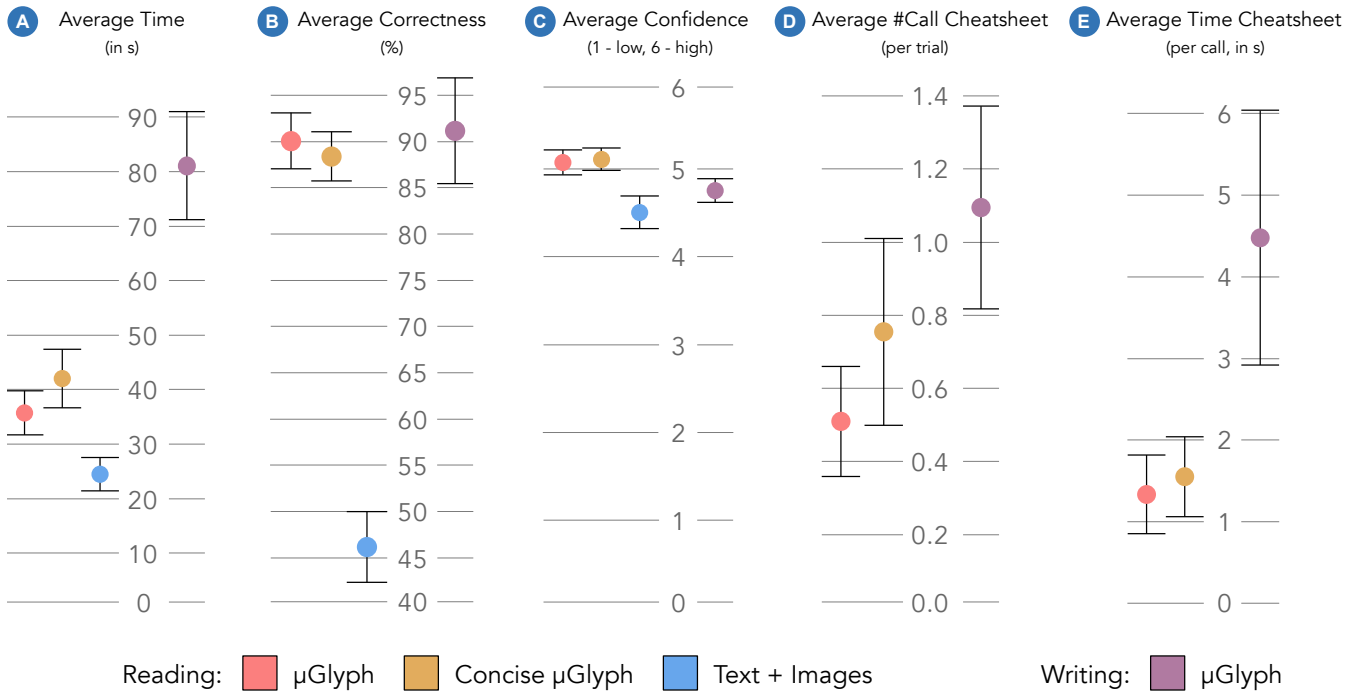


Figure 6: Overview of the study results with 95% confidence intervals. A) Average reading and writing time in seconds. B) Average percentage of correct reading and writing. C) Average confidence of the participants in their responses (1 - low confidence, 6 - high confidence). D) Average number of calls to the cheatsheet. E) Average time spent on cheatsheet for reading and writing.

- Symbol “p” interpreted as “palm” instead of “pinky” (3 / 39)
- Other types of one-off errors (8 / 39)

For text+images descriptions, out of the 97 errors made when reading 180 descriptions, the most common errors are:

- Incorrect start and end positions of the microgesture, e.g. a tap [7, 39] ($\overset{\circ}{\nabla}$) performed as $\overset{\circ}{\nabla}$; $\overset{\circ}{\nabla}$ (43 / 97)
- Incorrect or incomplete microgesture direction (11 / 97)
- A choice interpreted as a sequence (5 / 97)
- A press [8] ($\overset{\circ}{\nabla}$) performed as a press [38] ($\overset{\circ}{\nabla}$) and the opposite case (4 / 97)

Confidence: Participants were more confident in their responses after reading standard μGlyph (5 ± 0.1 , $n = 60$) and concise μGlyph (5 ± 0.1 , $n = 60$) than text+images descriptions (4.5 ± 0.2 , $n = 60$). There is a significant difference between text+images and μGlyph (POL: -2) as well as between text+images and concise μGlyph (POL: -2), but not between μGlyph and concise μGlyph (POL: 2).

Cheatsheet: On average, the cheatsheet was called more often with concise μGlyph (0.8 ± 0.5 , $n = 180$) than with standard μGlyph (0.5 ± 0.3 , $n = 180$). Participants spent on average $1.5s \pm 1.0$ reading the cheatsheet for concise μGlyph and $1.3s \pm 1.0$ for standard μGlyph. A more detailed analysis reveals that the cheatsheet has never been called on average for 37% of microgestures (corresponding to 11 out of 30 microgestures) when using standard μGlyph and for 20% of microgestures (6 out of 30 microgestures) when using concise μGlyph.

6.2 Writing using μGlyph

Time: On average, participants wrote a μGlyph expression in $1min 20s \pm 19s$. As expected, the more symbols there are to write and the less common the symbols are, for example l for left, the longer it takes to write a μGlyph expression. Indeed, the expressions with the least number of symbols, namely flexion, stretch, square and tap thumb with pinky, are, on average, written in less than 40s. The expressions with the highest number of symbols, namely tap and flap, linear finger to thumb slides and swipe index with middle and thumb, are, on average, written in over 1min 50s. The three other expressions, namely tap palm with index middle and ring finger, drag index or middle on stylus up/down, and pressing down on the lateral tip of the index, take between 1min 4s and 1min 44s to write.

Correctness: Using μGlyph, participants wrote the correct microgesture $91\% \pm 6$ of the time. There were 16 errors out of 180 μGlyph descriptions written. The most common errors are:

- Merged contexts, i.e. $\circ\circ$ written $\circ$ and $\bullet\bullet$ written $\bullet$ (5 / 16)
- Merged events, i.e. $\overset{\circ}{\nabla}$; $\overset{\circ}{\nabla}$ written $\overset{\circ}{\nabla}$ (2 / 16)
- Incorrect or missing movement direction (2 / 16)
- Parallel symbol “||” instead of choice symbol “|” (2 / 16)
- Incorrect symbol placement and missing contact glyph (2 / 16)

Confidence: Participants had a confidence in their responses of 4.8 ± 0.13 .

Cheatsheet: On average, per microgesture, participants called the cheatsheet 1.1 ± 0.6 times for a total of $4.5s \pm 3.1$ spent on it.

μGlyph forms of expression used: On average, participants used standard μGlyph 33% of the time and concise μGlyph 67% of the time. It varies considerably depending on the microgesture to be written. Indeed, Tap thumb with pinky, Tap palm with index, middle and ring and stretch are written in standard μGlyph by 8 participants (44%) and in concise μGlyph by 10 participants (56%). Linear finger-to-thumb slides, Flexion, Drag index or middle on stylus up or down are written in standard μGlyph by 3 participants (17%) and in concise μGlyph by 15 participants (83%).

7 DISCUSSION

In this section, we report the lessons learned from the conceptual and experimental study of μGlyph.

7.1 Improvement of the notation

The experimental results, for both reading and writing μGlyph, show errors in the interpretation of two symbols, namely choice “|” and parallel “||” symbols. We observed that the error of reading a choice symbol “|” as a sequence symbol “;” occurred only if the choice symbol was in long descriptions on a single line, but never occurred when it was placed at the beginning of a new line. Therefore, we recommend that μGlyph users spread long descriptions over several lines by cutting them off before a choice symbol.

The participants who misinterpreted a parallel symbol “||” as a choice symbol “|” explained to us that based on their knowledge of programming languages, the symbol “||” means to them a logical *or*. One way to solve this problem is to use a slanted parallel symbol: “//”.

7.2 μGlyph is usable

The experimental results show that μGlyph is usable by researchers and practitioners. Our participants first found μGlyph “frightening” [P3] and text+images “simpler to understand” [P4], but in the end, μGlyph is “comprehensible” [P3], “precise” [P4], and “simpler than text+images for complex microgestures” [P4]. P8, an HCI researcher, appreciated the high level of precision on the description of the microgesture. Moreover, P9, also an HCI researcher, pointed out that “having two symbols [“t” and “p”] with different meanings depending on their position, to the right or left of the glyph, requires mental attention”, yet, “μGlyph remains simple to understand”.

Like any notation, μGlyph requires some training to be mastered, even if “it looks intuitive” [P11]. As a concrete example of the need for training, we observed a common error during the training phase that never occurred during the experiment after the experimenter pointed it out to participants: many participants represented a thumb flexion using the symbol ▲ (since the thumb goes “up, toward the other fingers”) while the correct glyph is ▼ (since it goes “down, toward the palm”). This point should be highlighted in the μGlyph tutorial video. However, even though P6 thought that “[he] would need an hour of training”, based on the results of the experiment, participants still managed to read and write μGlyph correctly after a 9 min video tutorial and 10 min of writing practice.

It is unclear which form of expression is preferred by participants when reading μGlyph descriptions. While P7 and P15 found the concise μGlyph form “simpler”, “more logical” and “more visual” than the standard μGlyph form, P12 found the concise μGlyph form “not intuitive” and thought she would not remember its meaning in a week. However, for writing, 15 of the 18 participants, including those who preferred the standard μGlyph form, used the concise form to write microgestures requiring specification of a finger choice, i.e. the two microgestures Linear finger-to-thumb slides and drag index or middle finger on stylus up/down. Overall, participants thought that standard μGlyph is simpler to understand [P3, P4, P11, P18, P19] but becomes too verbose for microgestures with many features [P13, P18, P20]. Thus, they prefer concise μGlyph, called “expert form” by a participant [P11], to write and read “complex microgestures”, i.e. microgestures with choices of events and/or many features [P4, P6, P9, P10, P13, P14, P17, P18, P19]. We conclude that both forms of μGlyph expression are useful.

7.3 Literature is ambiguous

The experimental results confirm the terminological problems identified in the literature. Participants made several errors when reading descriptions from the literature and were not completely confident in their responses.

It was not uncommon for participants to be unsure of how to approach certain microgestures, such as “ream” or “tab”. The names did not give a clear idea of the corresponding microgestures. In these cases, participants relied solely on the associated images. For “ream” and “tab”, this led to respectively only 16% and 50% of correct answers. For press, however, the images did help participants to differentiate between  and , leading to 75% of correct answers. However, to avoid ambiguity, we recommend using the name “touch” for , as in [11, 47].

We also observed several errors due to a misunderstanding of the initial and final states of a microgesture, even if the textual description is precise. For instance, based on the precise description of a tap “pressing one part of the hand onto another part” [7] (), participants read the microgesture as ; , 61% of the time. Participants’ comments indicated that they were biased by their HCI expertise, particularly in touch interaction. P7, a computer science student, thought the name and description did not match, so he read the taps in a way that was familiar to him, i.e., a tap on a touchscreen ; . P11 reported that “in [his] experience [as an HCI research engineer], a tap is similar to a mouse click, whereas here it is similar to a mouse-down event”. Therefore, even if a microgesture is precisely described, its name can be misleading.

P16 was surprised that text+images descriptions were “less comprehensible than μGlyph”, specifically “to determine the actuator”. Additionally, during the writing part of the experiment, P13 noted that determining the actuator is difficult in the videos, especially for tapping since both the actuator and the receiver are moving.

These observations suggest that beyond precise descriptions as in μGlyph, the community needs to agree on names for classes of microgestures. Often, the name implies initial and final states in reference to the more established domains of touch interaction and 3D gesture interaction. As a first step towards this terminological goal,

we suggest the following names for the classes of microgestures described in μ Glyph at a high level of abstraction:

- Tab, Touch: $\overset{\circ}{\circ}$ (e.g. $\overset{\circ}{\nabla}$)
 - Tap: $\overset{\circ}{\circ}$; $\overset{\circ}{\circ}$ (e.g. $\overset{\circ}{\nabla}$; $\overset{\circ}{\blacktriangle}$)
 - Slide, Drag: $\overset{\circ}{\circ}$
 - Ream: $\overset{\circ}{\nabla}$; $\overset{\circ}{\blacktriangle}$ | $\overset{\circ}{\blacktriangle}$; $\overset{\circ}{\nabla}$ and extension to horizontal Ream: $\overset{\circ}{\nabla}$; $\overset{\circ}{\blacktriangle}$ | $\overset{\circ}{\blacktriangle}$; $\overset{\circ}{\nabla}$
 - Directional Swipe, Multi-finger snap, Drag&Drop: $\overset{\circ}{\nabla}$; $\overset{\circ}{\blacktriangle}$; $\overset{\circ}{\circ}$ | $\overset{\circ}{\nabla}$; $\overset{\circ}{\blacktriangle}$; $\overset{\circ}{\circ}$ | $\overset{\circ}{\nabla}$; $\overset{\circ}{\blacktriangle}$; $\overset{\circ}{\circ}$ | $\overset{\circ}{\nabla}$; $\overset{\circ}{\blacktriangle}$; $\overset{\circ}{\circ}$
- in other words $\overset{\circ}{\nabla}$; $\overset{\circ}{\blacktriangle}$; $\overset{\circ}{\circ}$

7.4 Rare properties

In building the database, we found very few examples of microgestures that would require the addition of textual symbols, such as those for phalanges, to μ Glyph. For instance, Whitmire et al. uses two-handed taps for text input [45]. For the standard μ Glyph, instead of introducing a new symbol, one can simply duplicate and mirror the graphical structure: ----- ----- . Another example would be to further differentiate fingertips to describe a tap on the very tip of the index by the thumb [8]. We purposefully decided not to increase the number of textual symbols in order not to overwhelm μ Glyph users with abbreviations for only one or two studied microgestures. Instead, we let users take advantage of the fact that μ Glyph is an hybrid notation, partly graphical and partly textual. This leaves the possibility of accurately describing new contact areas using words. For instance, $\overset{\circ}{\nabla}_{\text{left hand}}(\overset{\circ}{\bullet}_{\text{left hand}}); \overset{\circ}{\blacktriangle}_{\text{left hand}}$ describes a tap of the thumb of the left hand on the index finger of the left hand. Similarly, $\overset{\circ}{\nabla}_{\text{very tip}}(\overset{\circ}{\bullet}_{\text{very tip}}); \overset{\circ}{\blacktriangle}$ describes a tap of the thumb on the very tip of the index. This trade-off can be mitigated in the future: if the literature evolves, or if many studies make extensive use of previously understudied receivers, new textual symbols can be added to μ Glyph, as was done for other notations such as Labanotation [20]. However, no matter how many textual symbols are added, the core of the notation, i.e., movements, contexts, and event compositions, will remain the same, demonstrating a certain completeness and self-sufficiency of the μ Glyph notation.

8 ONLINE TOOL TO FACILITATE THE USE OF μ GLYPH

To promote the adoption of μ Glyph by researchers and practitioners, the proposed online website² offers three main services. Figure 7 shows a screenshot of these services.

First, we implemented a visual editor for μ Glyph descriptions with an export function to $\text{\LaTeX}$. We used this editor to translate the literature microgestures into μ Glyph and the $\text{\LaTeX}$ export feature to create the tables in Appendix B.

The website also contains the complete database, described above, along with a search engine allowing users to formulate targeted searches based on: 1) μ Glyph descriptions 2) keywords that match the names and descriptions of the microgestures as well as the descriptions of the grasps and objects studied in conjunction

with the microgestures. This search engine allows users to efficiently explore the database of existing microgestures, to relate and compare existing studies. To add a new paper, researchers can find our contact information on the website to send us their document and descriptions in μ Glyph.

Finally, to visually present a microgesture, the website includes a view of a 3D hand model using three.js and a tool to generate animations based on μ Glyph descriptions³. After inputting a μ Glyph description, users can watch an animation of the hand reproducing the microgesture. Each part of the μ Glyph description is used to create a keyframe of the animation. To play the animation, the keyframes are interpolated over a number of frames. Users can rotate the hand model to play the animation from any angle. Such a tool can be used to explain microgestures to users, similar to the Hololens 2 hand coach that teaches the pinch gesture [32]. In addition, users can export a SVG file containing all the animation keyframes. This feature was inspired from Esquisse [1], to help researchers easily export a graphical description of a microgesture with different hand orientations, as illustrated in Figure 8. The export function is implemented using the three SVG rendering plugin⁴. Usage scenarios of the tool are available in the accompanying video figure.

9 FUTURE RESEARCH BASED ON μ GLYPH

In this paper, we have presented μ Glyph, studied its expressiveness and usability, and presented tools that support the use of μ Glyph. While this is an important step toward a more rigorous hand microgesture field, there are clearly many avenues of research that this work does not directly address. Several of these are topics of current work by the authors. We outline these research avenues according to the main phases of the life cycle of an interactive system based on microgesture interaction: designing, developing, and evaluating.

9.1 Designing and developing microgesture interaction

Designing new microgestures. Using μ Glyph, it is possible to build a tree of all possible glyph combinations. By considering all the sequences of two μ Glyph events, without even taking the μ Glyph features into account, we already obtain more than 250 different microgestures. New types of microgestures, which have not yet been studied, can be identified. For instance, microgestures with momentum (e.g., the microgesture $\overset{\circ}{\blacktriangle}$; $\overset{\circ}{\nabla}$ is the momentum version of $\overset{\circ}{\nabla}$), or back and forth microgestures in the air (e.g., $\overset{\circ}{\nabla}$; $\overset{\circ}{\blacktriangle}$).

However, given the already wide variety of microgestures and the small number of human factor studies, we argue that at this time, the generation of new microgestures should be approached with caution. Nevertheless, the μ Glyph notation can help systematize the human factor studies.

Developing microgesture recognizers. One barrier for microgesture recognition is the training usually required [12, 37, 39] for a given set of microgestures. When a new microgesture is added to the set, beyond adding new sensors if required, new data from users

²<https://mic.imag.fr/microglyph/>

³The animation tool is a proof of concept. The tool is a work in progress and, at the time of writing, only works for some simple and concise μ Glyph expressions.

⁴<https://github.com/LokiResearch/three-svg-rendering>.

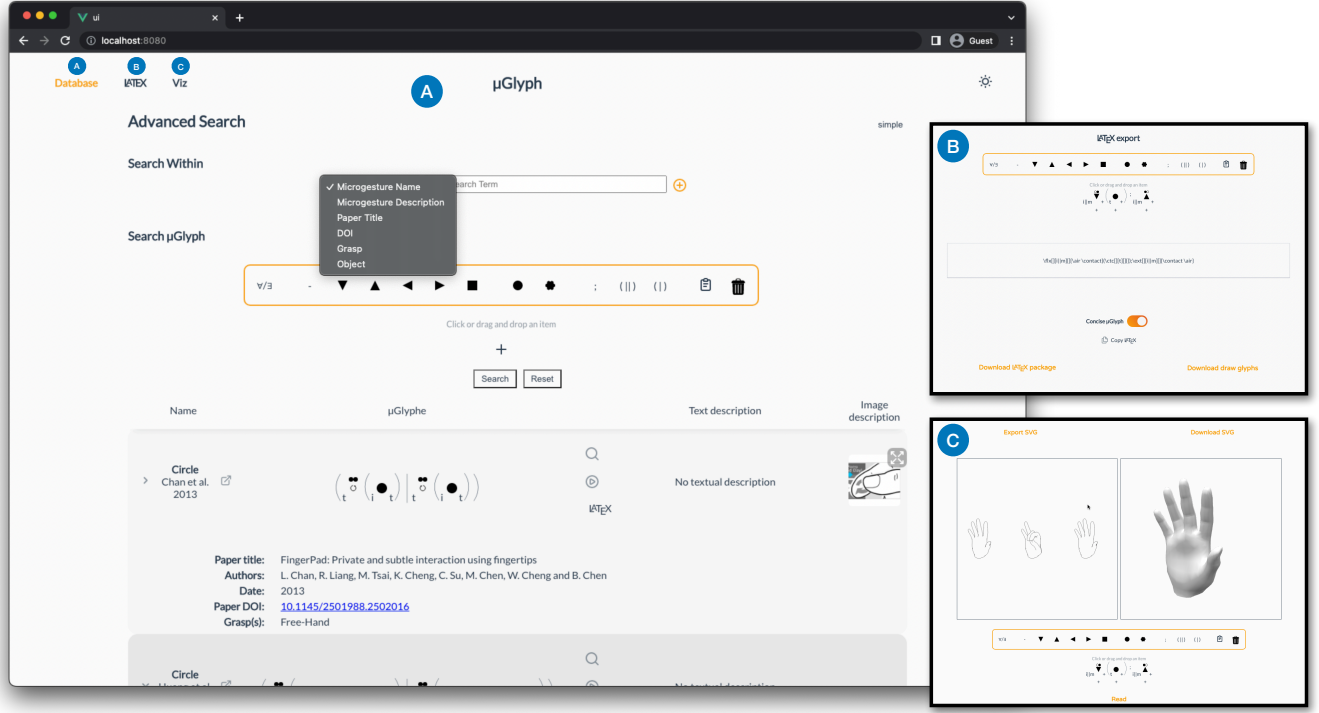


Figure 7: Screenshot of the μGlyph website. A) Database of microgestures: for each microgesture, name, description, paper and μGlyph description. B) μGlyph visual editor with $\LaTeX$ export. C) 3D animated hand generated from a μGlyph description and SVG export of the animation keyframes.

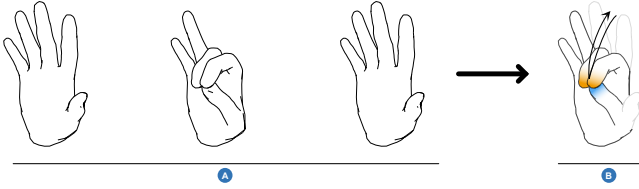


Figure 8: Creation of a graphical description of a microgesture using the SVG export. A) Keyframes exported from the 3D hand viewer (Figure 7 C). B) The graphical description created with the keyframes.

must be collected to re-train the model. One promising avenue is to create a classifier capable of recognizing elementary μGlyph events. The recognition of a new microgesture, composed of elementary μGlyph events, will then be straightforward, by combining the outputs from the classifier.

9.2 Evaluating microgesture interaction

Keystroke [6] or its extension for touch interaction [17, 35], makes it possible to accurately predict user performance. Such performance models do not yet exist for microgestures, but they are necessary for the field to develop further. We propose three approaches based on μGlyph to build such models

Using medical knowledge. Since the atomic movements of μGlyph are derived from biomechanics of the finger, i.e. extension, flexion, abduction and adduction, microgestures can be evaluated according to feasibility and/or human performance by relying on the medical literature, e.g. [2, 28, 29]. This may be even more important in defining inclusive microgesture sets that account for limitations due to specific conditions, such as finger arthritis. Although previous work has already used medical knowledge [24, 47], many questions related to the physical feasibility of microgestures remain. For instance, is it easier to perform a horizontal swipe, i.e. adduction or abduction, than a vertical swipe, i.e. flexion or extension.

Studying the memorability of microgesture sets. The size of microgesture sets varies considerably in the literature from three microgestures [31] to more than fifty [39]. But we have yet to understand how many microgestures a user can retain. μGlyph can be used as a basis for understanding which properties of a microgesture are easier to remember. For instance, it might be easier to remember a set of several taps (e.g. thumb-to-finger taps: $x \in \{i|m|r|p\}, y \in \{b|m|t\}, \ddot{\nabla}_t(\bullet_x \bullet_y); \ddot{\nabla}_t(\bullet_x \bullet_y)$), than a set of 5 different microgestures (e.g. tap: $\ddot{\nabla}_t; \ddot{\nabla}_t; \ddot{\nabla}_t$, press: $\ddot{\nabla}_t; \ddot{\nabla}_t; \ddot{\nabla}_t$, swipe up: $\ddot{\nabla}_t; \ddot{\nabla}_t; \ddot{\nabla}_t$, swipe down: $\ddot{\nabla}_t; \ddot{\nabla}_t; \ddot{\nabla}_t$ and stretch: $\ddot{\nabla}_t$).

Studying the proximity between microgestures. We have noted much confusion due to the fact that the microgestures are very close to each other, e.g. tap, touch, press, tap or swipe, slide, drag, drag and drop. μ Glyph could be used to assess how different two microgestures are, from the user's perspective. This could allow a designer to assign critical tasks to the most different microgestures and thus reduce the risk of false activation.

10 CONCLUSION

We have presented μ Glyph, a notation to unambiguously and precisely describe microgestures at different levels of abstraction. μ Glyph is based on a graphical and textual representation of the finger movements, the execution context, and optional features of the movements. μ Glyph aims to standardize microgesture descriptions by providing a convenient notation to researchers and practitioners. To evaluate the usability of the μ Glyph notation, we conducted an experiment with 18 participants. After a 9-minute course and a 10-minute training phase, our participants could accurately read μ Glyph 89% of the time and write correctly 91% of the time, as compared to 46% with conventional descriptions. To illustrate the descriptive power of the μ Glyph notation, we translated in μ Glyph a set of 118 microgestures from 17 literature papers. We gathered these translations, along with their original text and image descriptions, in an online database. To ease the adoption of the μ Glyph notation, we developed a set of tools for searching the database, writing and embedding μ Glyph expressions within $\LaTeX$ based research papers, as well as visualizing microgestures. We put forth that μ Glyph can serve as a common language amongst researchers and practitioners to advance the field of microgesture interaction.

ACKNOWLEDGMENTS

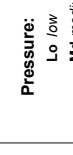
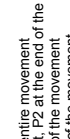
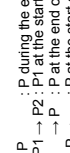
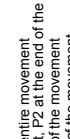
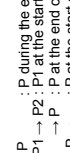
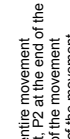
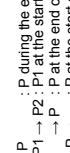
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The μ Glyph cheatsheet used during the experimental study.

Movement  Flexion ▼  Extension ▲  Adduction ►  Abduction ◄  Static ■	Contact  In the air ○  In contact ● ○ ● 2 dots: Final and starting positions are different ○ 1 dot: Final and starting positions are identical	Properties Pressure: Lo <i>low</i> Md <i>medium</i> Hi <i>high</i> Amplitude: ± % Cl <i>close</i> Ha <i>half</i> Op <i>open</i> Duration: ± s P : P during the entire movement P1 → P2 : P1 at the start, P2 at the end of the movement → P : P at the end of the movement P → : P at the start of the movement
Actuator  Standard  Concise  Standard  Concise m : middle i : index t : thumb r : ring p : pinky	Receiver  Standard  Concise  Standard  Concise n : nail t : tip m : middle b : base p : palm l : left r : right b : back f : front	Combination ;; sequence : choice : parallel Factorization Index or the middle tap $x \in \{ \text{index} \mid \text{middle} \}, \text{ }_x$ Index and middle touch on the palm Encapsulation Arrow replaced by the 2d drawing

B MICROGESTURES FROM LITERATURE PAPERS

The tables below show microgestures from literature papers. Each μGlyph description was exported from our website using the LaTeX export for μGlyph.

Table 8: First part of the set of microgestures derived from the pattern $\overset{\circ}{\bullet}; \overset{\circ}{\bullet}$.

Microgesture	Paper	μ Glyph
Click	Chan et al. 2013 [8]	$\overset{\circ}{\bullet}_t \nabla (\bullet_t); \overset{\circ}{\bullet}_t \blacktriangle$
Closed Gap touch	Way et al. 2014 [11]	$x0 \in \{t i m r p\},$ $x1 \in \{t i m r p\}, x0 \neq x1,$ $\overset{\circ}{\bullet}_{x0} \nabla (\bullet_{x1}) \overset{\circ}{\bullet}_{x1} \nabla (\bullet_{x0})$
Double Tap	Song et al. 2011 [40]	$\overset{\circ}{\bullet}_i \blacktriangle; \overset{\circ}{\bullet}_i \nabla (\bullet_i); \overset{\circ}{\bullet}_i \blacktriangle; \overset{\circ}{\bullet}_i \nabla (\bullet_i)$
External Surface Touch	Way et al. 2014 [11]	$\overset{\circ}{\bullet}_{t i m r p} \nabla (\bullet_i)$
Flick	Deyle et al. 2007 [12]	$x \in \{i m r p\}, \overset{\circ}{\bullet}_x \blacksquare_n (\bullet_t); \overset{\circ}{\bullet}_x \blacktriangle$
Finger Pinch	Way et al. 2014 [11]	$\overset{\circ}{\bullet}_t \nabla (\bullet_{i m r p} \bullet_{b m t})$
Finger Tap	Soliman et al. 2018 [39]	$\overset{\circ}{\bullet}_t \nabla (\bullet_{i m r} \bullet_{b m t})$
Fist Tap	Soliman et al. 2018 [39]	$\overset{\circ}{\bullet}_t \nabla (\bullet_i) \overset{\circ}{\bullet}_{i m r p} \blacksquare$
Make a fist	Chan et al. 2016 [7]	$\overset{\circ}{\bullet}_{t i m r p} \nabla \rightarrow cl$
Mult Flicks	Lee et al. 2020 [30]	$x \in \{i m m r r p i m r m r p i m r p\},$ $\overset{\circ}{\bullet}_x \blacksquare_n (\bullet_t); \overset{\circ}{\bullet}_x \blacktriangle$
Mult Taps	Lee et al. 2020 [30]	$x \in \{i m m r r p i m r m r p i m r p\},$ $y \in \{\text{nail center} \text{nail tip}\},$ $\overset{\circ}{\bullet}_t \nabla (\bullet_x \bullet_y); \overset{\circ}{\bullet}_t \blacktriangle$
Pinch gesture	Saponas et al. 2009 [36]	$\overset{\circ}{\bullet}_t \nabla (\bullet_{i m r p} \bullet_t) \overset{\circ}{\bullet}_t \blacktriangle$
Press on tip of index	Chan et al. 2013 [8]	$\overset{\circ}{\bullet}_t \nabla (\bullet_i \bullet_t)$
Snip	Wolf et al. 2011 [47]	$(\overset{\circ}{\bullet}_i \blacktriangleright (\bullet_m) \overset{\circ}{\bullet}_m \blacktriangleleft (\bullet_i)); (\overset{\circ}{\bullet}_i \blacktriangleleft \overset{\circ}{\bullet}_m \blacktriangleright)$
Stretch	Sharma et al. 2019 [38]	$\overset{\circ}{\bullet}_{r p} \blacktriangle (\bullet_i)$
Stretch	Sharma et al. 2019 [38]	$\overset{\circ}{\bullet}_{m r p} \blacktriangle (\bullet_p)$

Table 9: Second part of the set of microgestures derived from the pattern $\overset{\circ}{\bullet}$; $\overset{\circ}{\blacktriangle}$.

Microgesture	Paper	μGlyph
Tab	Wolf et al. 2011 [47]	$\overset{\circ}{\blacktriangledown}_{i m r p}(\bullet_t)$
Tab	Wolf et al. 2011 [47]	$\overset{\circ}{\blacktriangledown}_{i m r i m}(\bullet)$
Tab index or middle on object	Wolf et al. 2011 [47]	$x \in \{i m\}, y \in \{\}, \overset{\circ}{\blacktriangledown}_x(\bullet); \overset{\circ}{\blacktriangle}_x$
Tap	Huang et al. 2016 [24]	$\overset{\circ}{\blacktriangledown}_t(\bullet_{i m r p} \bullet_{b m t})$
Tap	Lee et al. 2020 [30]	$\overset{\circ}{\blacktriangledown}_t(\bullet_{i m r p} \bullet_{\text{nail tip nail root nail inner nail outer nail center}}); \overset{\circ}{\blacktriangle}_t$
Tap	Sharma et al. 2019 [38]	$\overset{\circ}{\blacktriangledown}_t(\bullet); \overset{\circ}{\blacktriangle}_t$
Tap	Sharma et al. 2019 [38]	$\overset{\circ}{\blacktriangledown}_m(\bullet); \overset{\circ}{\blacktriangle}_m$
Tap	Sharma et al. 2019 [38]	$\overset{\circ}{\blacktriangledown}_i(\bullet); \overset{\circ}{\blacktriangle}_i$
Tap	Sharma et al. 2021 [37]	$\overset{\circ}{\blacktriangledown}_{t i m}(\bullet); \overset{\circ}{\blacktriangle}_{t i m}$
Tap	Whitmire et al. 2017 [45]	$\overset{\circ}{\blacktriangledown}_t(\bullet_{i m r p} \bullet_{b m t}); \overset{\circ}{\blacktriangle}_t$
Tap all fingers together	Chan et al. 2016 [7]	$\overset{\circ}{\blacktriangledown}_{t i i m r i p}(\bullet)$
Tap and Flap	Soliman et al. 2018 [39]	$\overset{\circ}{\blacktriangledown}_t(\bullet_{i \bullet_{b m t}}) \left(\overset{\circ}{\blacktriangledown}_{m i r p} \mid \overset{\circ}{\blacktriangle}_{m i r p} \right)_{\substack{op \rightarrow cl \\ cl \rightarrow op}}$
Tap index and middle finger together	Chan et al. 2016 [7]	$\overset{\circ}{\blacktriangle}_i(\bullet_m) \overset{\circ}{\blacktriangle}_m(\bullet_i)$
Tap of the thumb on other fingers phalanx and nail	Wambecke et al. 2021 [44]	$\left(\overset{\circ}{\blacktriangledown}_t(\bullet_{i m} \bullet_{b m t}) \overset{\circ}{\blacktriangledown}_t(\bullet_{i \bullet_{t,i}}) \overset{\circ}{\blacktriangledown}_t(\bullet_{r \bullet_{n}}) \right); \overset{\circ}{\blacktriangle}_t$
Tap palm with index, middle and ring fingers	Chan et al. 2016 [7]	$\overset{\circ}{\blacktriangledown}_{i m r}(\bullet_p)$
Tap palm with thumb	Chan et al. 2016 [7]	$\overset{\circ}{\blacktriangledown}_t(\bullet_p)$
Tap the wheel	Wolf et al. 2011 [47]	$x \in \{t i m r p\}_x \overset{\circ}{\blacktriangledown}_x; \overset{\circ}{\blacktriangle}_x$
Tap thumb and pinky	Chan et al. 2016 [7]	$\overset{\circ}{\blacktriangledown}_p(\bullet_t)$
Tap thumb with middle finger	Chan et al. 2016 [7]	$\overset{\circ}{\blacktriangledown}_m(\bullet_t)$
Tap thumb with index and middle finger	Chan et al. 2016 [7]	$\overset{\circ}{\blacktriangledown}_{i m}(\bullet_t)$
Tap thumb with index finger	Chan et al. 2016 [7]	$\overset{\circ}{\blacktriangledown}_i(\bullet_t)$
Tap thumb with ring and pinky finger	Chan et al. 2016 [7]	$\overset{\circ}{\blacktriangledown}_{r p}(\bullet_t)$
Tap thumb with ring finger	Chan et al. 2016 [7]	$\overset{\circ}{\blacktriangledown}_r(\bullet_t)$
Tap to the very tip	Chan et al. 2013 [8]	$\overset{\circ}{\blacktriangledown}_t(\bullet_{\text{very tip}}); \overset{\circ}{\blacktriangle}_t$
Thumb Up	Wolf et al. 2011 [47]	$\overset{\circ}{\blacktriangle}_t$
Touch	Wolf et al. 2011 [47]	$\overset{\circ}{\blacktriangledown}_t(\bullet_{i m r p} \bullet_n)$

Table 10: Microgestures using pressure.

Microgesture	Paper	μ Glyph
Force Tap	Whitmire et al. 2017 [45]	$x \in \{i m r p\}, y \in \{b m t\},$
Press	Saponas et al. 2009 [36]	
Press	Sharma et al. 2019 [38]	
Press	Sharma et al. 2019 [38]	
Press	Sharma et al. 2019 [38]	
Press	Wolf et al. 2011 [47]	
Press	Yi et al. 2019 [48]	
Pull	Saponas et al. 2009 [36]	

Table 11: First part of the set of microgestures derived from the pattern $\ddot{\bullet}$.

Microgesture	Paper	μGlyph
Drag Index or Middle finger on stylus up or down	Wolf et al. 2011 [47]	$_{i m} \ddot{\nabla}(\bullet) _{i m} \ddot{\blacktriangle}(\bullet)$
Drag and Drop Index on Thumb	Wolf et al. 2011 [47]	$_i \ddot{\circ}(\bullet); _i \ddot{\nabla}(\bullet); _i \ddot{\circ}$
Drag and Drop Middle on Index	Wolf et al. 2011 [47]	$_m \ddot{\circ}(\bullet_d); _m \ddot{\nabla}(\bullet_d); _m \ddot{\circ}$
Drag and Drop Thumb on Index-side	Wolf et al. 2011 [47]	$x \in \{i m r p\}, _t \ddot{\circ}(\bullet_x \bullet_n); _t \ddot{\blacktriangleright}(\bullet_x \bullet_n); _t \ddot{\circ}$
Drag fingers around the wheel	Wolf et al. 2011 [47]	$_{i m r p} \ddot{\nabla}(\bullet)$
Drag middle finger above object	Wolf et al. 2011 [47]	$_m \ddot{\blacktriangle}(\bullet)$
Drag thumb along object	Wolf et al. 2011 [47]	$_t \ddot{\blacktriangleleft}(\bullet) _t \ddot{\blacktriangleright}(\bullet)$
Drag thumb around object	Wolf et al. 2011 [47]	$_t \ddot{\nabla}(\bullet) _t \ddot{\blacktriangle}(\bullet)$
Extension	Sharma et al. 2021 [37]	$_{t i m} \ddot{\blacktriangle}(\bullet)$
Fingertip slide	Soliman et al. 2018 [39]	$_t \ddot{\blacktriangleleft}(\bullet_{i m r} \bullet_n) _t \ddot{\blacktriangleright}(\bullet_{i m r} \bullet_n)$
Flexion	Sharma et al. 2021 [37]	$_{t i m} \ddot{\nabla}(\bullet)$
Flip	Wolf et al. 2011 [47]	$_{i m r p} \ddot{\blacktriangle}(\bullet_t); _{i m r p} \ddot{\blacktriangle}$
HSwipe	Lee et al. 2020 [30]	$x \in \{i m r p\}, y \in \{\text{nail top} \text{nail center} \text{nail root}\},$ $_t \ddot{\circ}(\bullet_x \bullet_y); \left(_t \ddot{\blacktriangleleft}(\bullet_x \bullet_y) _t \ddot{\blacktriangleright}(\bullet_x \bullet_y) \right); _t \ddot{\circ}$
Linear Finger to Thumb slide	Soliman et al. 2018 [39]	$_{i m r} \ddot{\nabla}(\bullet_t) _{i m r} \ddot{\blacktriangle}(\bullet_t)$
Linear Thumb to Finger slide	Soliman et al. 2018 [39]	$_t \ddot{\nabla}(\bullet_{i m r} \bullet_{l r}) _t \ddot{\blacktriangle}(\bullet_{i m r} \bullet_{l r})$
Move	Chan et al. 2016 [7]	$_t \ddot{\circ}(\bullet_t)$
Mult HSwipe	Lee et al. 2020 [30]	$x \in \{i m m r r p i m r m r p i m r p\},$ $y \in \{\text{nail tip} \text{nail center} \text{nail root}\},$ $_t \ddot{\circ}(\bullet_x \bullet_y); \left(_t \ddot{\blacktriangleleft}(\bullet_x \bullet_y) _t \ddot{\blacktriangleright}(\bullet_x \bullet_y) \right); _t \ddot{\circ}$
Multi-finger snap	Deyle et al. 2007 [12]	$_t \ddot{\circ}(\bullet_p); _t \ddot{\blacktriangleleft}(\bullet_r); _t \ddot{\blacktriangleleft}(\bullet_m); _t \ddot{\blacktriangleleft}(\bullet_i); _t \ddot{\circ}$
Ream	Wolf et al. 2011 [47]	$_{i m r p} \ddot{\nabla}(\bullet_t) _{i m r p} \ddot{\blacktriangle}(\bullet_t)$
Rotational thumb-to-finger slide	Soliman et al. 2018 [39]	$_t \ddot{\blacktriangleleft}(\bullet_{i m r} \bullet_{b m}) _t \ddot{\blacktriangleright}(\bullet_{i m r} \bullet_{b m})$
Slide	Whitmire et al. 2017 [45]	$x \in \{i m r p\}, y \in \{b m t\},$ $_t \ddot{\circ}(\bullet_x); \left(_t \ddot{\nabla}(\bullet_x) _t \ddot{\blacktriangle}(\bullet_x) \right); _t \ddot{\circ}$

Table 12: Second part of the set of microgestures derived from the pattern $\ddot{\circ}$.

Microgesture	Paper	μ Glyph
Swipe	Sharma et al. 2019 [38]	$\overset{\ddot{\circ}}{t} \blacktriangleleft (\bullet) \overset{\ddot{\circ}}{t} \blacktriangleright (\bullet)$
Swipe	Sharma et al. 2019 [38]	$\overset{\ddot{\circ}}{t} \blacktriangledown (\bullet); \overset{\ddot{\circ}}{t} \blacktriangle (\bullet)$
Swipe	Sharma et al. 2019 [38]	$\overset{\ddot{\circ}}{m} \blacktriangleleft (\bullet) \overset{\ddot{\circ}}{m} \blacktriangleright (\bullet)$
Swipe	Sharma et al. 2019 [38]	$\overset{\ddot{\circ}}{m} \blacktriangledown (\bullet) \overset{\ddot{\circ}}{m} \blacktriangle (\bullet)$
Swipe	Sharma et al. 2019 [38]	$\overset{\ddot{\circ}}{i} \blacktriangleleft (\bullet) \overset{\ddot{\circ}}{i} \blacktriangleright (\bullet)$
Swipe	Sharma et al. 2019 [38]	$\overset{\ddot{\circ}}{i} \blacktriangledown (\bullet) \overset{\ddot{\circ}}{i} \blacktriangle (\bullet)$
Swipe	Song et al. 2011 [40]	$\overset{\ddot{\circ}}{i t}$
Swipe Left	Sharma et al. 2021 [37]	$\overset{\ddot{\circ}}{t i m} \blacktriangleleft (\bullet)$
Swipe Right	Sharma et al. 2021 [37]	$\overset{\ddot{\circ}}{t i m} \blacktriangleright (\bullet)$
Swipe across all fingers with thumb	Chan et al. 2016 [7]	$\overset{\circ}{t} (\bullet); \overset{\ddot{\circ}}{t} \blacktriangleleft (\bullet); \overset{\ddot{\circ}}{t} \blacktriangleleft (\bullet); \overset{\ddot{\circ}}{t} \blacktriangleleft (\bullet); \overset{\circ}{t}$
Swipe index and middle fingers with thumb	Chan et al. 2016 [7]	$\overset{\circ}{t} (\bullet); \left(\overset{\ddot{\circ}}{t} \blacktriangledown (\bullet) \overset{\ddot{\circ}}{t} \blacktriangle (\bullet) \right); \overset{\circ}{t}$
Swipe index finger with thumb	Chan et al. 2016 [7]	$\overset{\circ}{t} (\bullet); \left(\overset{\ddot{\circ}}{t} \blacktriangledown (\bullet) \overset{\ddot{\circ}}{t} \blacktriangle (\bullet) \right); \overset{\circ}{t}$
Swipe middle and ring fingers with thumb	Chan et al. 2016 [7]	$\overset{\circ}{t} (\bullet); \left(\overset{\ddot{\circ}}{t} \blacktriangledown (\bullet) \overset{\ddot{\circ}}{t} \blacktriangle (\bullet) \right); \overset{\circ}{t}$
Swipe middle finger with thumb	Chan et al. 2016 [7]	$\overset{\circ}{t} (\bullet); \left(\overset{\ddot{\circ}}{t} \blacktriangledown (\bullet) \overset{\ddot{\circ}}{t} \blacktriangle (\bullet) \right); \overset{\circ}{t}$
Swipe thumb leftward	Chan et al. 2013 [8]	$\overset{\ddot{\circ}}{t} \blacktriangleleft (\bullet)$
Thumb index rub backward	Deyle et al. 2007 [12]	$\overset{\ddot{\circ}}{t} \blacktriangleleft (\bullet)$
V Swipe	Lee et al. 2020 [30]	$x \in \{i m r p\},$ $y \in \{\text{nail inner} \text{nail center} \text{nail outer}\},$ $\overset{\circ}{t} (\bullet); \left(\overset{\ddot{\circ}}{t} \blacktriangledown (\bullet) \overset{\ddot{\circ}}{t} \blacktriangle (\bullet) \right); \overset{\circ}{t}$

Table 13: Microgestures derived from the pattern $\overset{\circ\circ}{-}$.

Microgesture	Paper	μGlyph
Down	Endres et al. 2011 [14]	$\overset{\circ\circ}{i} \blacktriangledown$
Finger Curl	Way et al. 2014 [11]	$\overset{\circ\circ}{t j m r p} \blacktriangledown$
Finger Extend	Way et al. 2014 [11]	$\overset{\circ\circ}{t j m r p} \blacktriangle \text{ MCP}$
Finger Lift	Way et al. 2014 [11]	$\overset{\circ\circ}{t j m r p} \blacktriangledown \text{ MCP}$
Left	Endres et al. 2011 [14]	$\overset{\circ\circ}{i} \blacktriangleleft$
Open Palm	Hauslschmid et al. 2015 [22]	$\overset{\circ\circ}{t j i m n r p} \blacktriangle$
Right	Endres et al. 2011 [14]	$\overset{\circ\circ}{i} \blacktriangleright$
Stretch	Sharma et al. 2019 [38]	$\overset{\circ\circ}{r p} \blacktriangle$
Swipe	Hauslschmid et al. 2015 [22]	$\overset{\circ\circ}{i} \blacktriangleright \overset{\circ\circ}{i} \blacktriangleleft$
Up	Endres et al. 2011 [14]	$\overset{\circ\circ}{i} \blacktriangle$

Table 14: Drawing microgestures.

Microgesture	Paper	μGlyph
Clock wise and counter clockwise circle	Hauslschmid et al. 2015 [22]	$\overset{\circ}{\circ} \overset{\circ}{\circ}$
Circle	Chan et al. 2013 [8]	$\overset{\circ}{\circ}_t(\bullet_t) _t\overset{\circ}{\circ}(\bullet_t)$
Circle	Huang et al. 2016 [24]	$\overset{\circ}{\circ}_t(\bullet_{i m r p} \bullet_{b m t}) _t\overset{\circ}{\circ}(\bullet_{i m r p} \bullet_{b m t})$
Circle	Sharma et al. 2021 [37]	$\overset{\circ}{\circ}_{t i m}(\bullet)$
Circle CW	Endres et al. 2011 [14]	$\overset{\circ}{\circ}_i$
Circle CCW	Endres et al. 2011 [14]	$\overset{\circ}{\circ}_i$
Circle CW & CCW	Wolf et al. 2011 [47]	$x \in \{i m r p\}, \overset{\circ}{\circ}_t(\bullet_x) _t\overset{\circ}{\circ}(\bullet_x)$
Circle sidewise	Wolf et al. 2011 [47]	$\overset{\circ}{\circ}_t(\bullet_{i m r p}) _t\overset{\circ}{\circ}_t(\bullet_{i m r p})$
Circle thumb over side of index finger	Chan et al. 2016 [7]	$\overset{\circ}{\circ}_t(\bullet)$
Draw Circle	Soliman et al. 2018 [39]	$\overset{\circ}{\circ}_t(\bullet_b); \overset{\circ}{\circ}_t(\bullet_{i m r p})$
Square	Huang et al. 2016 [24]	$\overset{\circ}{\circ}_t(\bullet_{i m r p} \bullet_{b m t}) _t\overset{\circ}{\circ}_t(\bullet_{i m r p} \bullet_{b m t})$
Square CW	Endres et al. 2011 [14]	$\overset{\circ}{\circ}_i$
Square CCW	Endres et al. 2011 [14]	$\overset{\circ}{\circ}_i$
Triangle	Huang et al. 2016 [24]	$\overset{\circ}{\circ}_t(\bullet_{i m r p} \bullet_{b m t}) _t\overset{\circ}{\circ}_t(\bullet_{i m r p} \bullet_{b m t})$
Triangle CW	Endres et al. 2011 [14]	$\overset{\circ}{\circ}_i$
Triangle CCW	Endres et al. 2011 [14]	$\overset{\circ}{\circ}_i$
Zigzag	Sharma et al. 2021 [37]	$\overset{\circ}{\circ}_{t i m}(\bullet)$